Department of Computer Science And Engineering (CSE)
BRAC University

CSE 251 – Electronic Devices and Circuits

Lecture: 03

Operational Amplifier (OP-AMP)

Closed Loop Configuration

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Feedback

Issue with Open Loop Configuration:

- The gain (**A**) of an ideal op amp in Open Loop Config is infinity (∞), practically extremely large ($\mathbf{10^5}$).
- The power supply ($+V_{sat}$ and $-V_{sat}$) limits the op amp's output.
- Thus, a small voltage difference (V_d) in the Input Terminals can quickly reach the **Saturation.**

What We Need:

- We require a method to have a **finite gain** in the Op-Amp or a mechanism to **change the gain** as per the usage requirements in the practical settings.

How to Achieve:

- Introduce **Negative Feedback** in the Op-Amp circuit configuration.
- This method feeds a **portion** of ***output*** (V_o) back to the ***inverting input*** terminal (V_-).
- Idea: the output (V_o) will become stable due to a self-correcting mechanism.

Types of Feedback

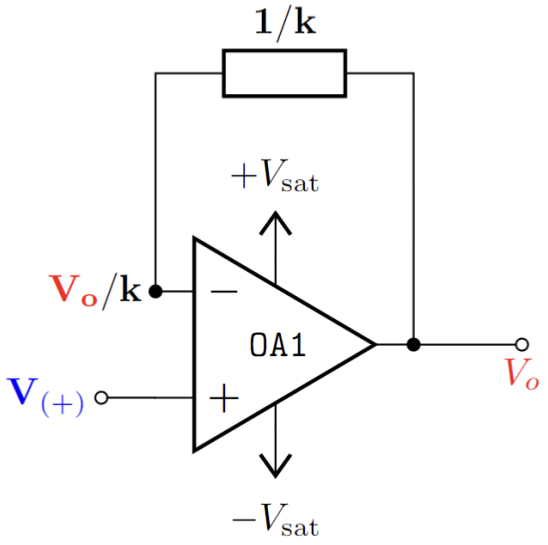
1. Negative Feedback:

Output voltage is fed to the inputs **Negatively.**

The output voltage (V_o) is connected to the **inverting** terminal (V_-).

$V_O \uparrow \Rightarrow \frac{V_O}{k} \uparrow \Rightarrow V_{(-)} \uparrow$ So, $V_d \downarrow = V_{(+)} - V_{(-)} \uparrow$ So, $V_O \downarrow \propto V_d$

Hence, V_o **is controlled, not allowed to increase rapidly.**



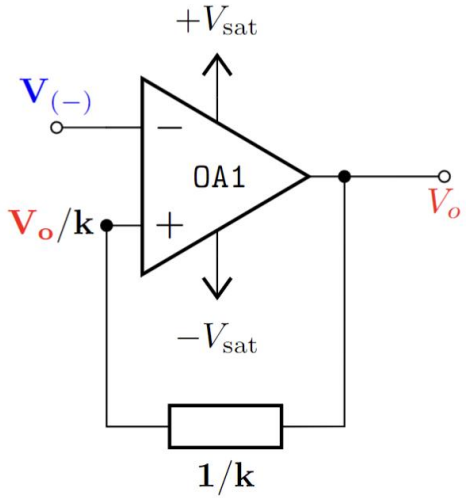
2. Positive Feedback:

Output voltage is fed to the inputs **Positively.**

The output voltage (V_o) is connected to the **non-inverting** terminal (V_+).

$V_O \uparrow \Rightarrow \frac{V_O}{k} \uparrow \Rightarrow V_{(+)} \uparrow$ So, $V_d \uparrow = V_{(+)} \uparrow - V_{(-)}$ So, $V_O \uparrow \propto V_d$

Hence, **no control over V_o, it increases indefinitely.**



Negative Feedback

The output voltage (V_o) is connected to the **inverting** terminal (V_-).

Hence, $V_- = \frac{V_o}{k}$

We know, $V_o = AV_d$

Or, $V_o = A(V_+ - V_-)$

$$= A\left(V_+ - \frac{V_o}{k}\right)$$

$$= AV_+ - \frac{A}{k}V_o$$

Or, $V_o\left(1 + \frac{A}{k}\right) = AV_+$

So, $\frac{V_o}{V_+} = \frac{A}{1 + \frac{A}{k}} = \frac{1}{\frac{1}{A} + \frac{1}{k}}$

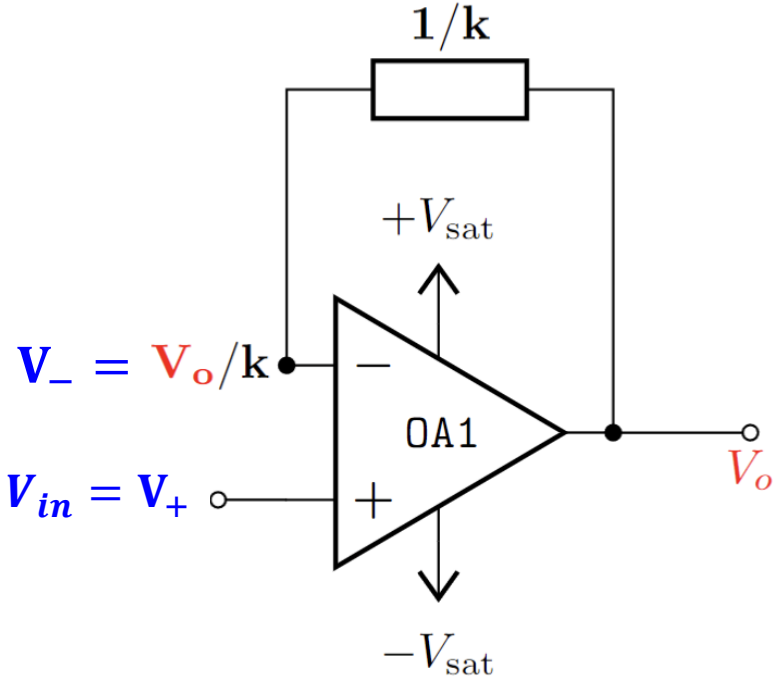
For ideal case, $A \to \infty$

Then, $\frac{1}{A} \to 0$.

$$\therefore \frac{V_o}{V_+} = k$$

Here, V_+ is the Input to the Op-Amp.

$$\therefore \frac{V_o}{V_{in}} = \frac{\text{Output}}{\text{Input}} = k = Gain$$



This is the new Effective Amplification Gain (k) or, the **Close Loop Gain**.

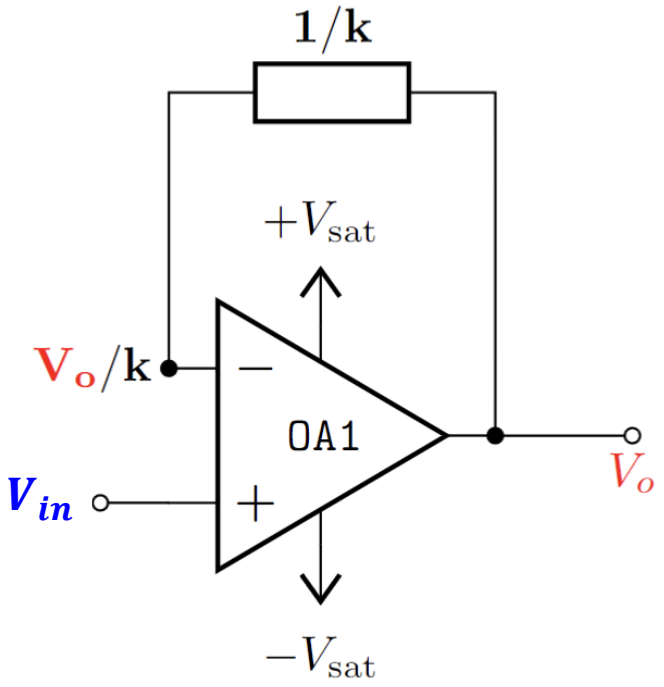
Negative Feedback

We get using Negative Feedback,

$$\frac{V_O}{V_{in}} = k$$

Or, $V_o = kV_{in}$

- We can control this gain 'k' by defining what portion of the V_o is to be fed back to the V_-.
- If 'k' = **10** (meaning we are feeding $(1/k)th = \left(\frac{1}{10}\right)th$ of the V_o back to the V_-), we will get $V_o = 10 \times V_{in}$. that is 10-fold gain.
- This is how Negative Feedback allows us to control and design the Close Loop Gain (k), as per we need.
- Positive Feedback doesn't provide this privilege to control the Gain. Hence, we shall not study Positive Feedback.

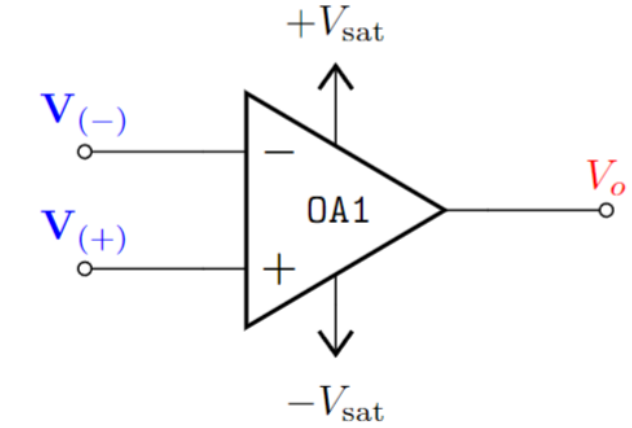


Negative Feedback

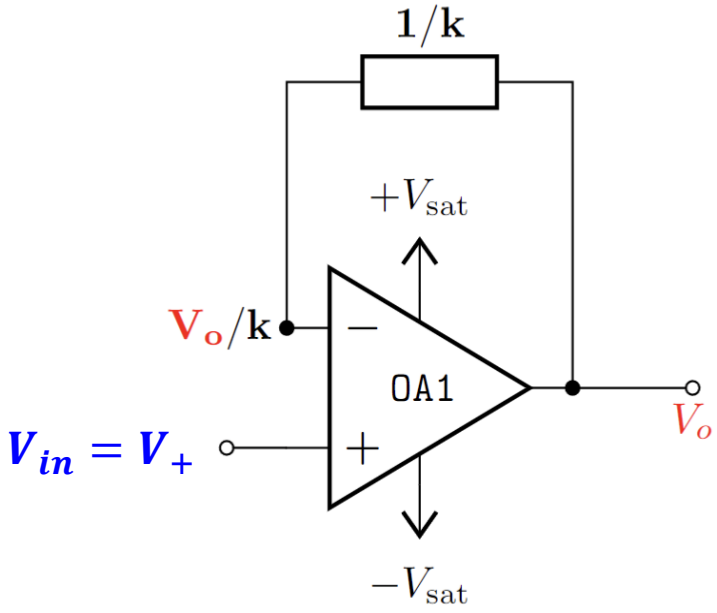
Open Loop vs. Closed Loop

Open Loop (OL) Configuration

$$V_d = V_{(+)} - V_{(-)}$$



Closed Loop (CL) Configuration with "Negative Feedback"



Property	Open Loop (OL)	Closed Loop (CL)
Input	$V_d = V_{(+)} - V_{(-)}$	$V_{in} = V_+$
Gain	$\frac{V_O}{V_d} = A$: Fixed, ideally ∞	$\frac{V_O}{V_+} = k \ll A$: Controllable
Usage	Comparator	Linear Amplifier

Negative Feedback

In negative feedback, the V_O is transformed as:

$$V_{(-)} = \frac{1}{k} \cdot V_O$$

How to design this feedback factor $(1/k)$?

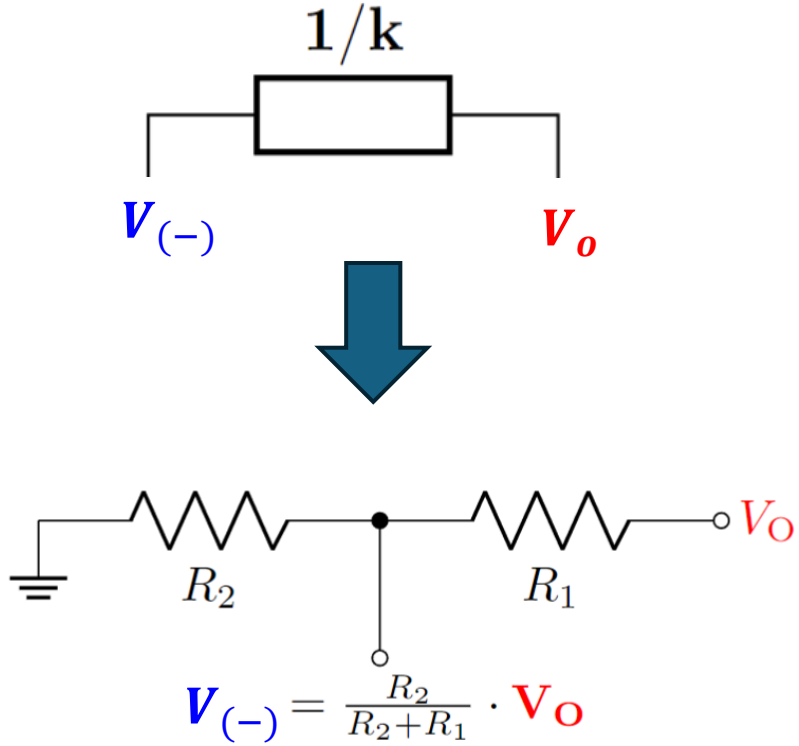
It can be achieved with a ***Voltage Divider Network.***

In the given network, we can write,

$$\frac{1}{k} = \frac{R_2}{R_1 + R_2}$$

For $k = 5$, we can set $R_1 = 4k\Omega$ and $R_2 = 1k\Omega$.

This is how a voltage divider network can act as a **multiplier/factor** in the **feedback** branch.



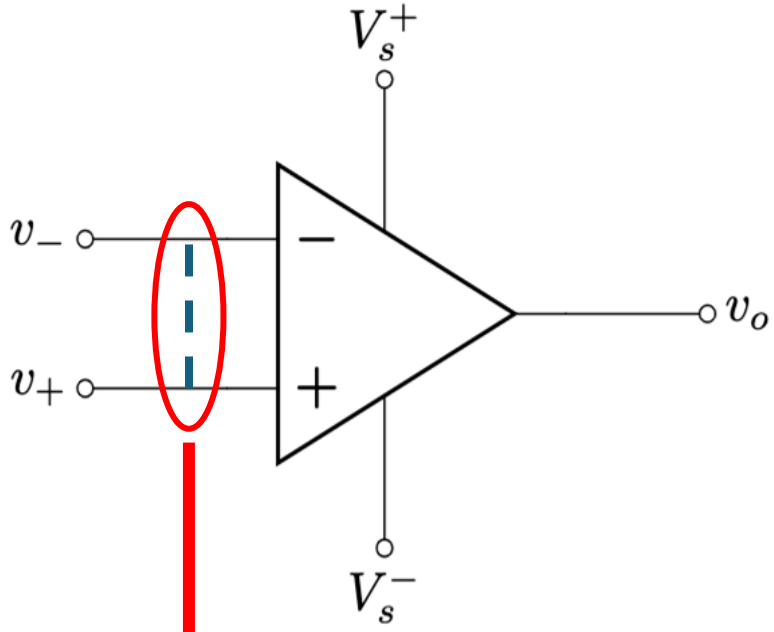
Voltage Divider Network acting as Feedback Line

Virtual Short

This is an important property of the Op-Amp under **Negative Feedback**.

Remember for "**Ideal**" op-amp,

- **Infinite** input resistance, $\boldsymbol{R_i = \infty}$ = open circuit
- **Zero** output resistance, $\boldsymbol{R_o = 0}$ = short circuit
- $\boldsymbol{i_i = 0}$ and $\boldsymbol{i_+ = 0}$.



When there is negative feedback:

In an ideal op-amp, "$\boldsymbol{A}$" (or $\boldsymbol{K}$) is **infinitely high**. Thus,

for a finite output voltage $\boldsymbol{v_o}$:

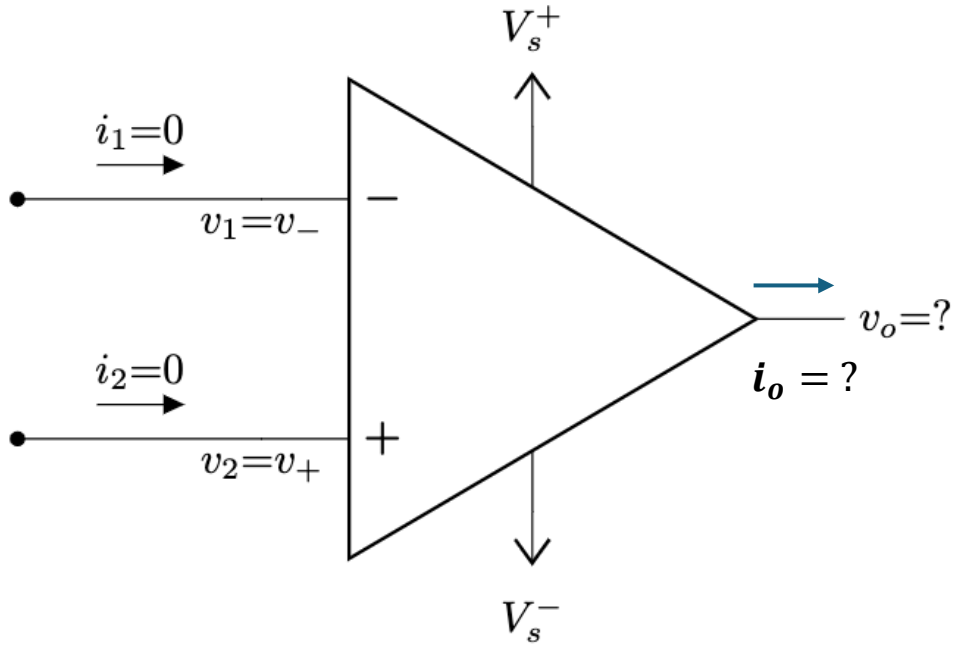
$$\frac{V_o}{A} = V_d \to 0 \Rightarrow V_+ = V_-.$$

This is called **virtual short circuit.**

There is no direct physical connection between V_- and V_+, still their voltages are equal.

Solving Op-Amp in Closed Loop

While solving Closed Loop Op-Amp Circuits, Consider **Ideal Op-Amp** approximation. We shall analyze only the **Negative Feedback** configuration.



Two Main Rules to Follow:

1. Virtual Shorting: $V_+ = V_-$
2. Zero input bias current: $i_- = i_+ = 0$

General Steps to Follow for Most Common Circuits:

1. First, look at the V_+ node. If it is grounded, then $V_+ = 0V$.
2. If V_+ is not grounded, apply KCL at V_+ node to find it's value (consider $i_+ = 0$).
3. Apply **Virtual Short** to find V_-, which is exactly equal to V_+ found in previous steps.
4. Apply KCL at the V_- node, where $i_- = 0$.
5. If there are extra (multiple) nodes in the feedback line, apply KCL to every extra nodes.
6. Solve the KCL equations to find V_o. Check if V_o is within the Saturation limit.
7. For output current i_o, apply KCL at the V_o node.

Example 01

Solve the Op-Amp circuit to find V_o

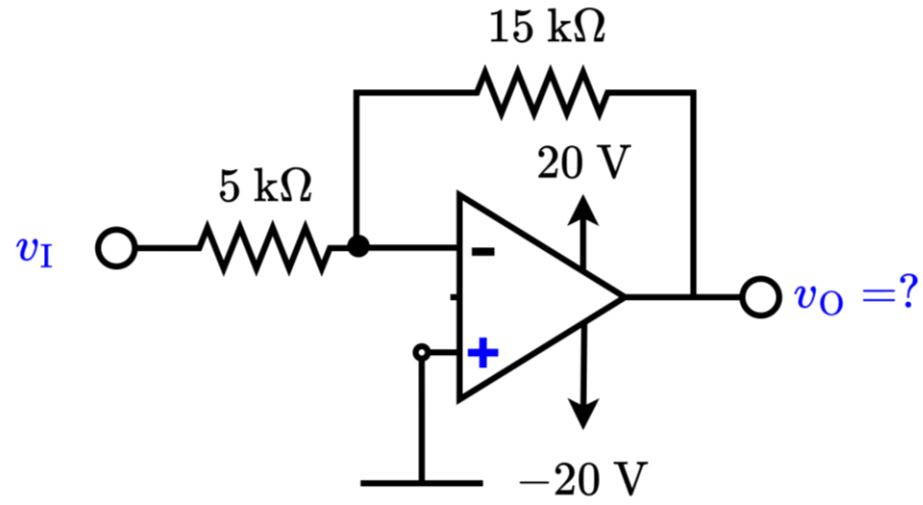
1. Since V_+ is connected to ground,
$$V_+ = 0\ \text{V}$$
2. Since there is negative feedback,
Using Virtual Short, $V_- = V_+ = 0\ \text{V}$
3. Apply KCL at V_- node,

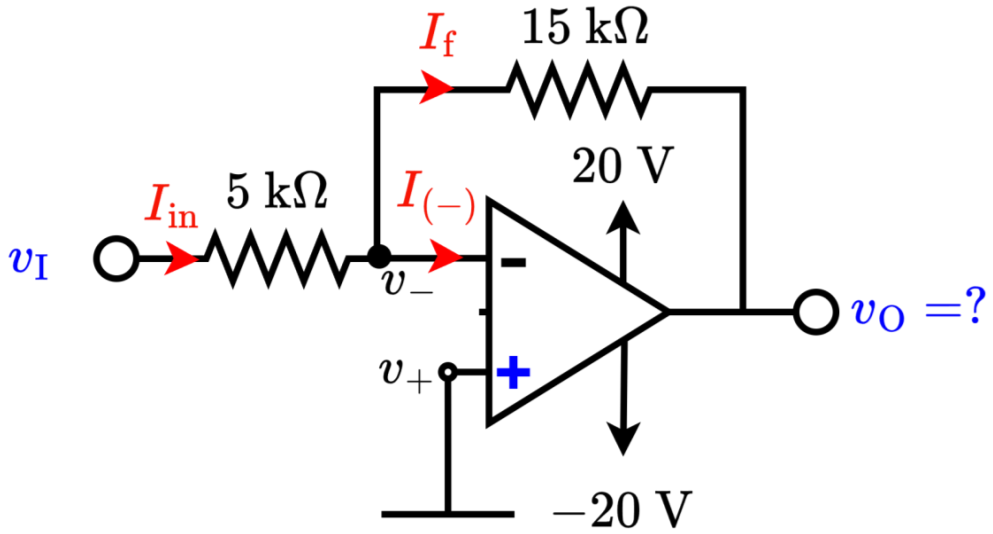
$$I_{in} = I_{(-)} + I_f$$

$$\Rightarrow \frac{V_I - 0}{5} = 0 + \frac{0 - V_o}{15}$$

$$\Rightarrow \frac{V_o}{V_I} = -\frac{15}{5} = -3$$

$$\Rightarrow V_o = -3V_I$$





Example 01

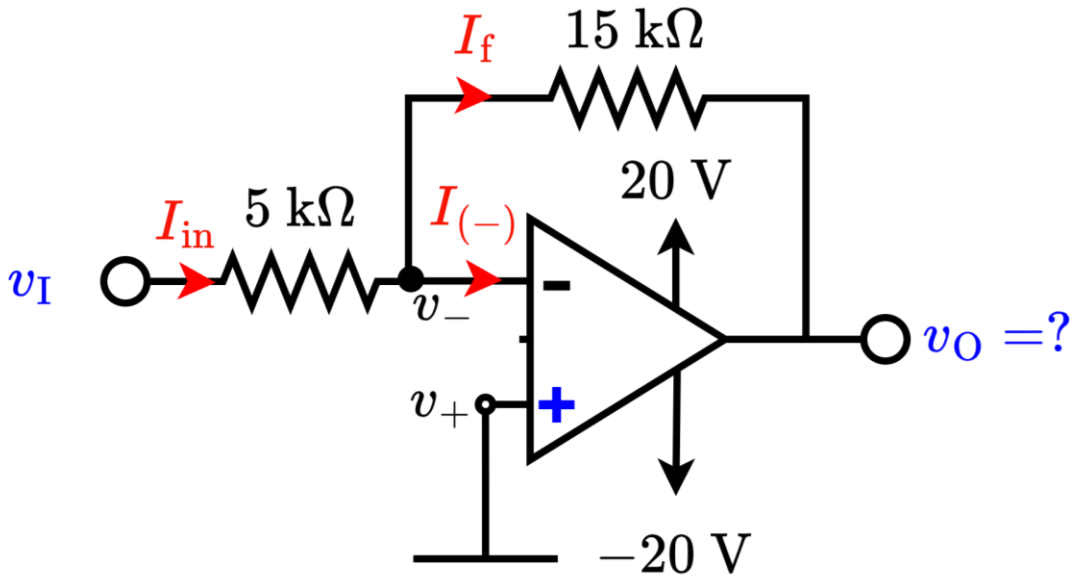
So, the equation becomes,

$$V_o = -3V_I$$

But can V_o have any values? No, it is always limited by the Saturation Voltage Limits.

If, V_I = 5V, $V_o = -3 \times 5 = -15V$
Since, $-20V < -15V < 20V$
So, $V_o = -15V$

If, V_I = -8V, $V_o = -3 \times -8 = 24V$
Since, $24V > 20V$, crossed the limit
So, $V_o = 20V$



A fundamental Last Step: You must check if V_o is within the saturation limit or not. Then, use the value of V_o for determining other circuit values, such as I_o.

Closed Loop Op-Amp Configurations

There are several specific circuit configurations under closed loop that are used to implement certain mathematical operations.

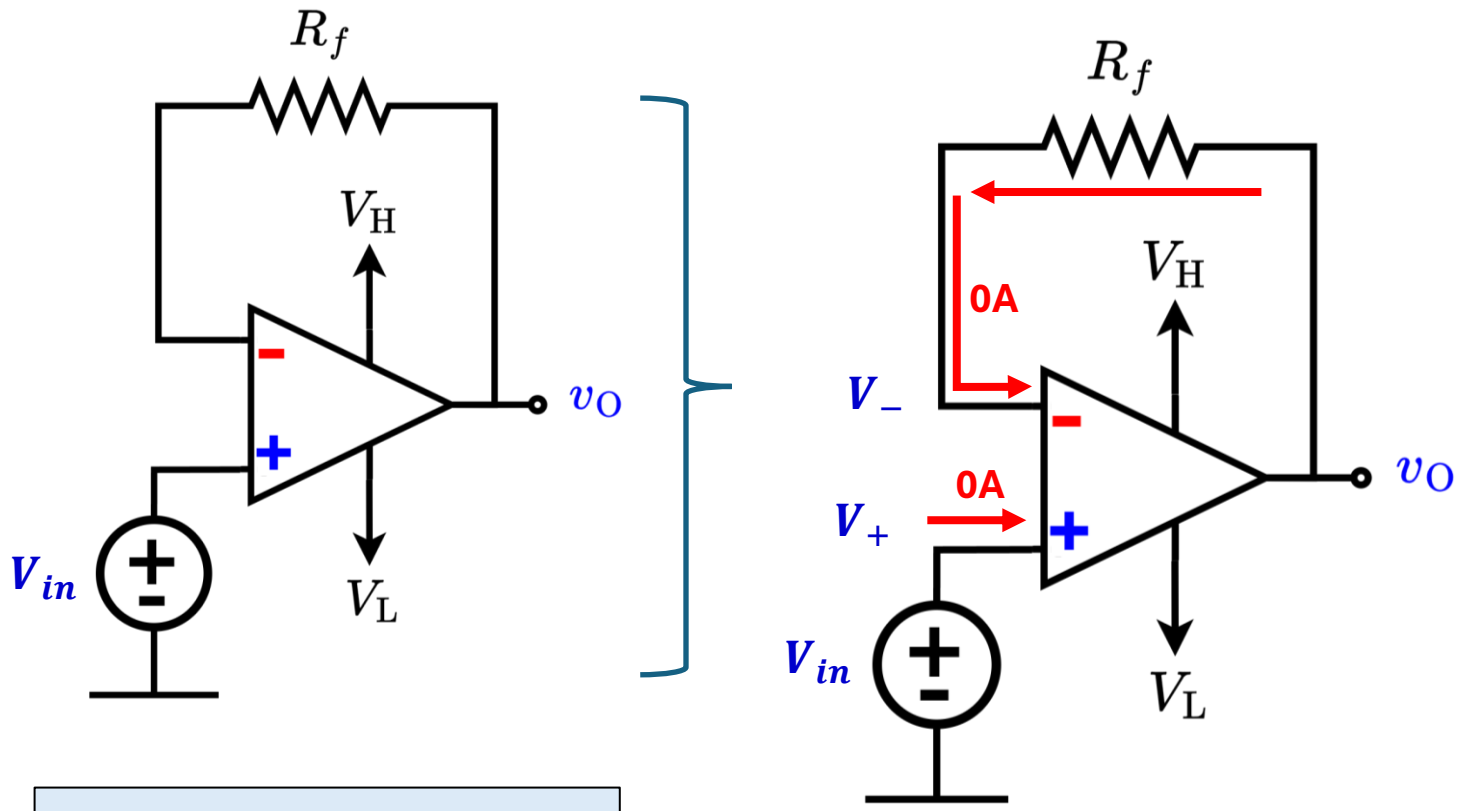
1. Voltage Follower
2. Inverting Amplifier
3. Weighted Summer (Inverting)
4. Non-Inverting Amplifier
5. Weighted Subtractor
6. Integrator (Inverting)
7. Differentiator (Inverting)
8. Exponential Converter
9. Logarithmic Converter
10. Multiplier
11. Divider

We shall apply "Ideal Op-Amp" approximation to analyze these circuit configuration, their working principles, and derive their equations.

These will be analyzed after we study Diode.

Voltage Follower

A circuit whose Output V_o follows the Input V_{in} (equal).



Voltage Follower Circuit

Input voltage V_I is applied to the V_+ node.

$$V_+ = V_{in}$$

Using Virtual Short,

$$V_- = V_+ = V_{in}$$

In ideal Op-Amp,

$$I_- = 0$$

$$\Rightarrow \frac{V_0 - V_-}{R_f} = 0$$

$$\Rightarrow V_0 = V_-$$

$$\Rightarrow V_0 = V_{in}$$

Since, Output is equal to Input, this circuit is also called **Buffer**. Here, the value of R_f doesn't matter.

Voltage Follower

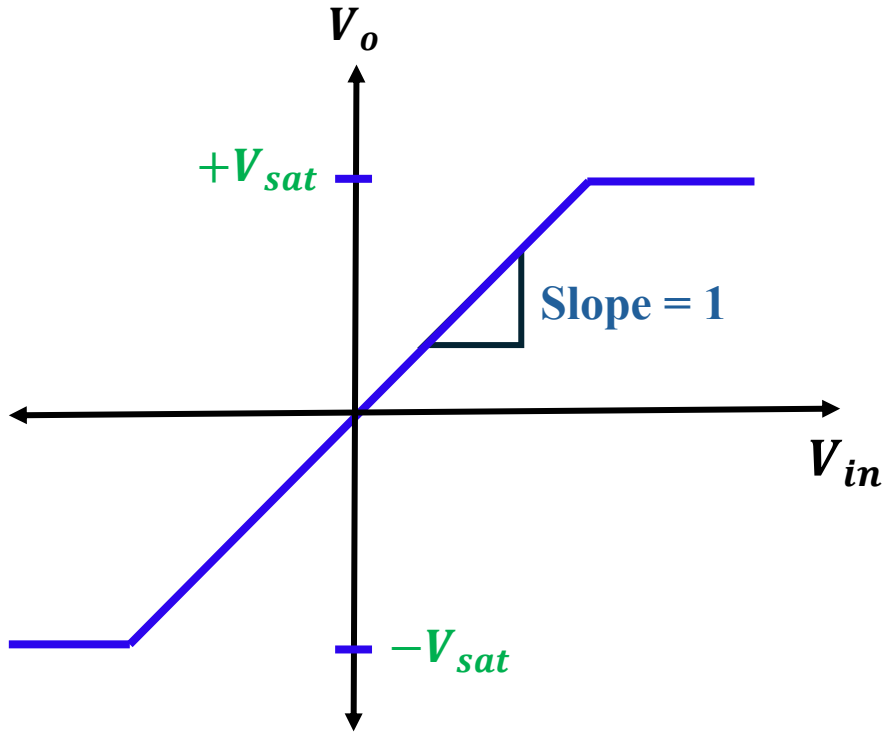
Since, $V_0 = V_{in}$

$$Gain = \frac{V_o}{V_{in}} = 1$$

In the **VTC** plot for Voltage follower, there is a straight line with **slope = 1**.

Purpose of Voltage Follower / Buffer?

Although it has unity gain, no amplification or attenuation, the primary purpose to use a buffer is for resistance matching, featuring very high input resistance and low output resistance to isolate high-resistance sources from low-resistance loads, thereby preventing loading effects.



VTC of Voltage Follower

Inverting Amplifier

A circuit that not only amplifies the Input V_{in} to the Output V_o as well as inverts (sign change).

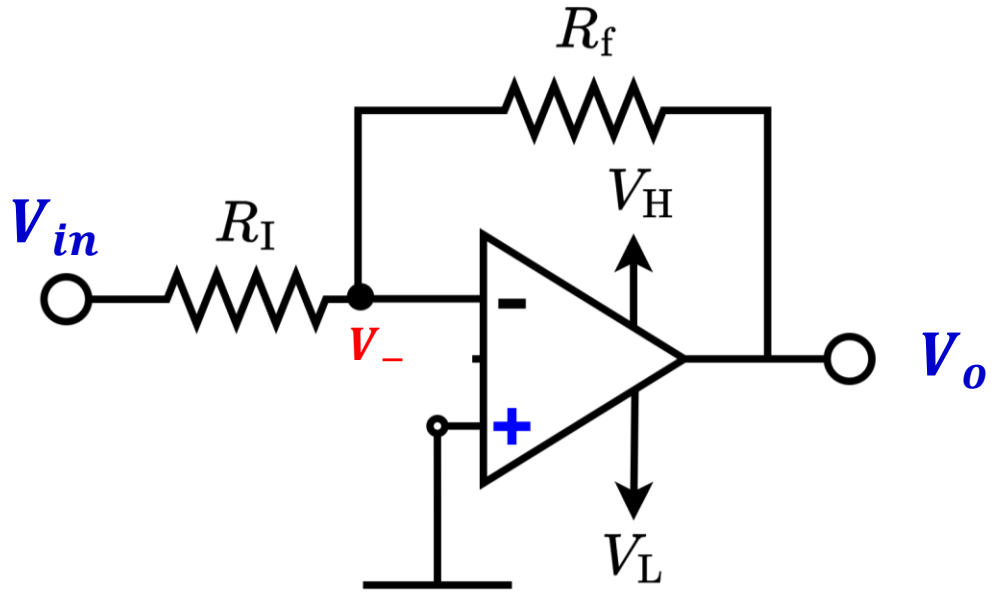
The V_+ node is grounded, $V_+ = 0V$
Using Virtual Short, $V_- = V_+ = 0V$

Applying KCL at V_- node,

$$\frac{V_{in} - V_-}{R_I} = I_- + \frac{V_- - V_o}{R_f}$$

$$\Rightarrow \frac{V_{in}}{R_I} = -\frac{V_o}{R_f} \quad [I_- = 0]$$

$$\Rightarrow \boldsymbol{V_o = -\frac{R_I}{R_f} \times V_{in}}$$



Comparing the final equation with,
$V_o = Gain \times V_{in}$

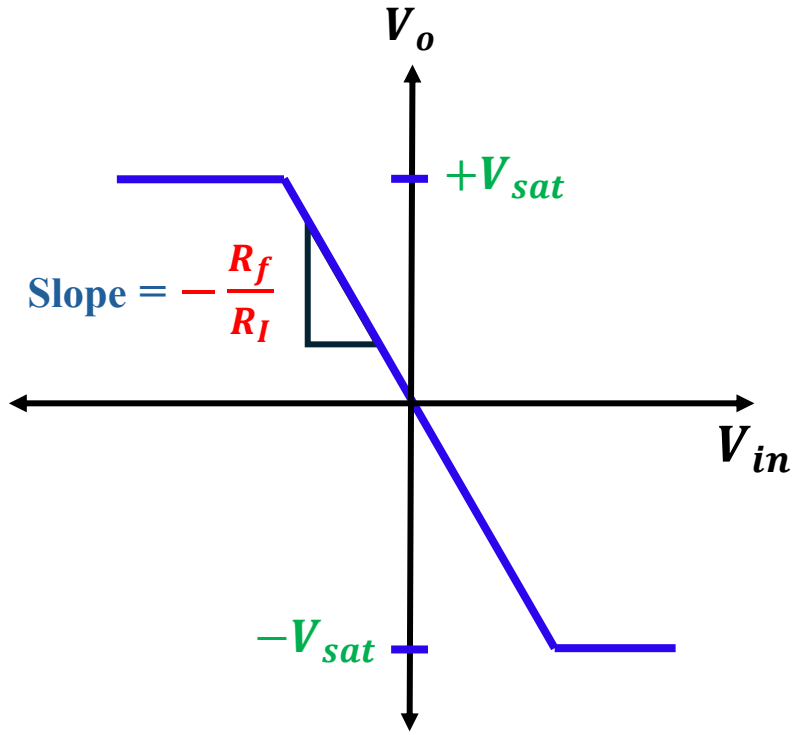
→ $Gain = -\frac{R_f}{R_I}$ →

Gain is **negative**, which depends on the values of R_f and R_I

Inverting Amplifier

- Gain Equation of an Inverting Amplifier $= -\frac{R_f}{R_I}$
- Hence, the **VTC** plot is a straight line with a **Negative slope.**
- Due to the negative gain, Output has opposite sign of Input (**Inverted Output**).
- We can change the **Gain** by changing of the values of R_f and R_I, which enables us to adjust the gain as per our requirements.
- However, V_o can never cross the saturation limit.

$$-V_{sat} \leq V_o \leq +V_{sat}$$



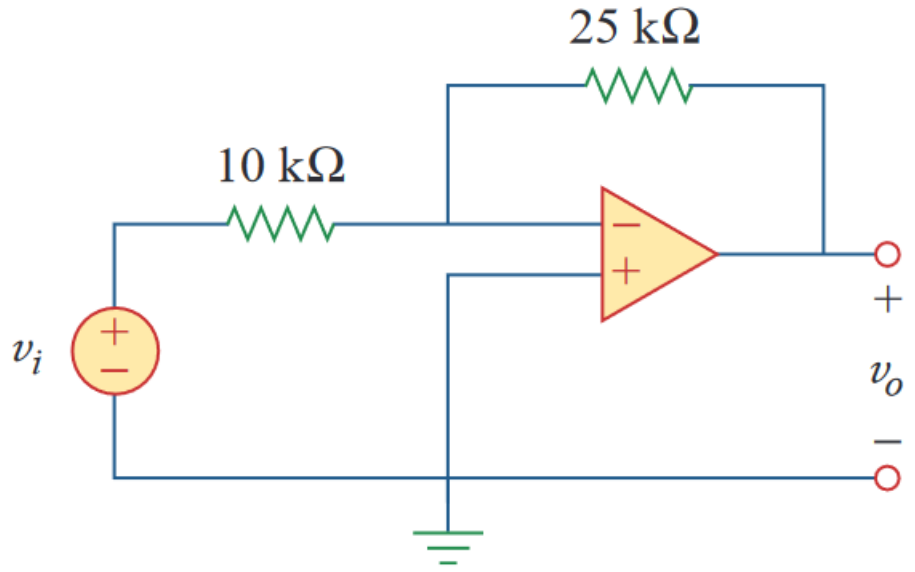
VTC of an Inverting Amplifier

Example 02

If $\boldsymbol{v_i = 0.5}$ V, calculate:

(a) Output voltage $\boldsymbol{v_o}$.

(b) Current in the **10 kΩ** resistor.



(a) This is an inverting amplifier.

$$\boldsymbol{v_o = -\frac{R_f}{R_i} \cdot v_i = -2.5 v_i = -1.25\ \text{V}}$$

(b) Current through the **10 kΩ** resistor is

$$i = \frac{v_i - v_-}{R_i} = \frac{0.5 - 0}{10}\ \text{mA} = \boldsymbol{50\ \mu\text{A}}$$

Example 03

Design the circuit such that the closed loop voltage gain is $-\mathbf{5}$. Assume the op-amp is driven by an ideal sinusoidal source, $\boldsymbol{v_I = 0.1 \sin(\omega t)\ (V)}$, that can supply a maximum current of $\mathbf{5\ \mu A}$.

For an Inverting Amplifier,

$$Gain = -\frac{R_f}{R_i} = -5$$

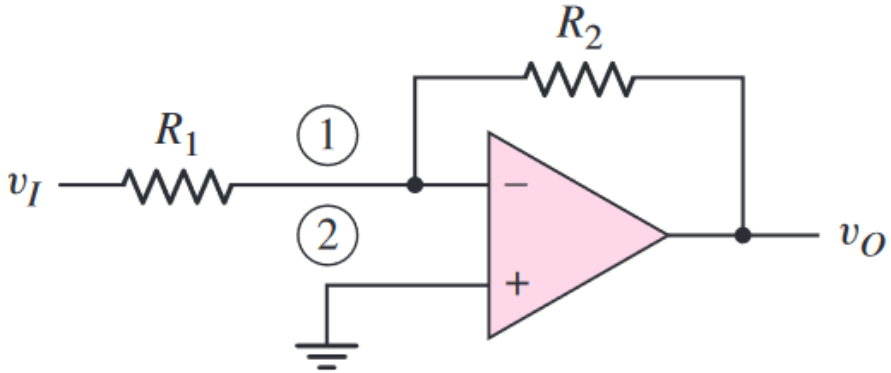
$$\Longrightarrow \boldsymbol{R_f = 5R_i}$$

The supply current from the source is basically the current following through $\boldsymbol{R_1}$

$$i = \frac{v_I - v_-}{R_1} = \frac{0.1\sin(\omega t)}{R_1}$$

Given, $\boldsymbol{i_{max} = 5\mu = \frac{0.1}{R_1}}$

Hence, $\boldsymbol{R_1 = 20k}$; and $\boldsymbol{R_f = 5 \times 20k = 100k}$



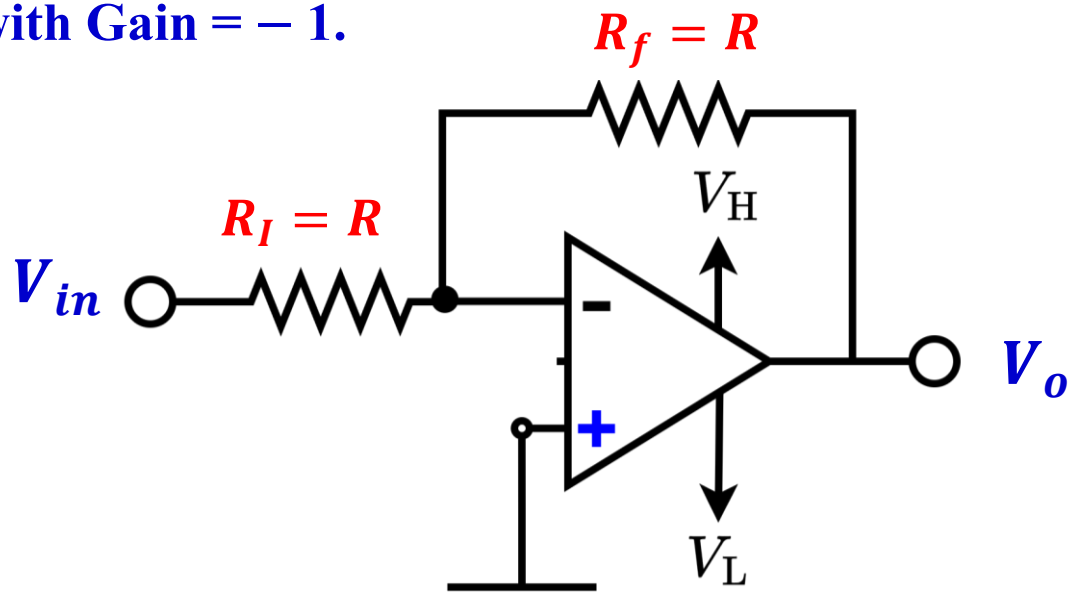
Sign Changer

A circuit that only changes the sign of the Input V_{in}, that is $V_o = -V_{in}$. No Amplification, just change the sign.

This can be realized easily using an **Inverting Amplifier with Gain = − 1.**

$$Gain = -\frac{R_f}{R_I} = -\frac{R}{R} = -1$$

$$V_o = -1 \times V_{in}$$



This circuit acts like a 180° phase shifter (Phase Inverter) or "Inverting Buffer", which is used frequently in analog electronics where polarity reversal is required.

Weighted Inverting Summer

A circuit that can add **multiple** weighted Inputs and **inverts** the summation in the Output.

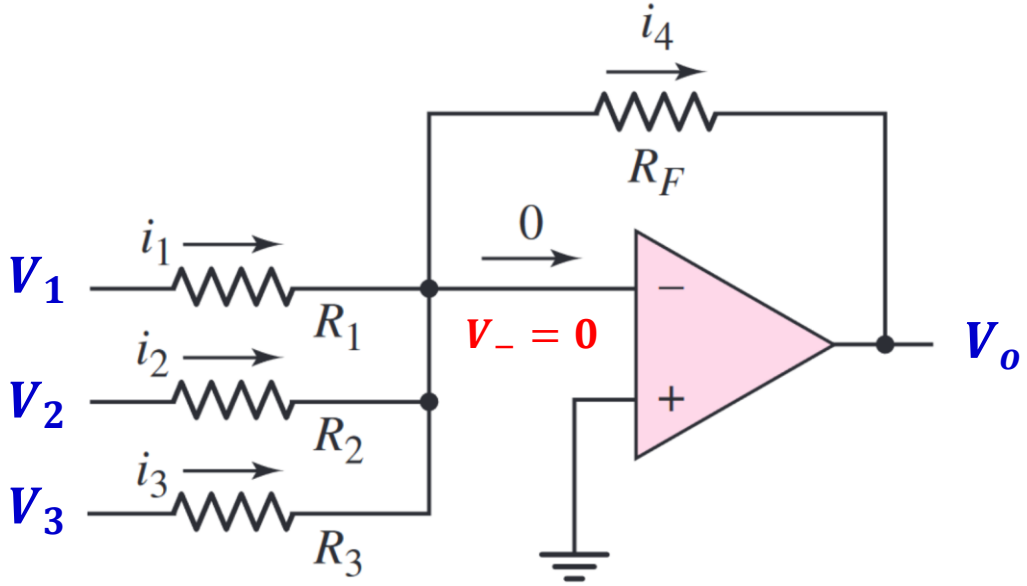
There are three Inputs: $\boldsymbol{V_1, V_2, V_3}$
The $\boldsymbol{V_+}$ node is grounded, $\boldsymbol{V_+ = 0V}$
Using Virtual Short, $\boldsymbol{V_- = V_+ = 0V}$

Applying KCL at $\boldsymbol{V_-}$ node,

$$\frac{V_1 - V_-}{R_1} + \frac{V_2 - V_-}{R_2} + \frac{V_3 - V_-}{R_3} = I_- + \frac{V_- - V_o}{R_f}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_o}{R_f}$$

$$\Rightarrow \boldsymbol{V_o = -\left(\frac{R_f}{R_1} \times V_1 + \frac{R_f}{R_2} \times V_2 + \frac{R_f}{R_3} \times V_3\right)}$$



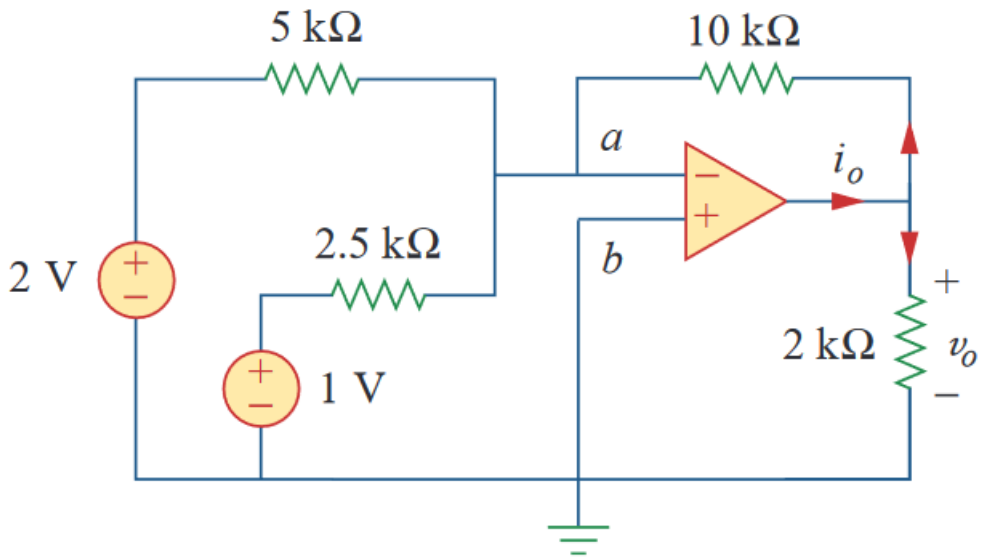
Each input is multiplied by a **factor**, the weighted products are added, finally the sum is **inverted**.

Example 04

Calculate:

(a) Output voltage $\boldsymbol{v_o}$.

(b) Output current $\boldsymbol{i_o}$.



(a) This is a weighted inverting summer.

$$\boldsymbol{v_o = -\left(\frac{R_f}{R_1} \cdot V_1 + \frac{R_f}{R_2} \cdot V_2\right)}$$

$$\boldsymbol{= -\left(\frac{10}{5} \cdot 2 + \frac{10}{2.5} \cdot 1\right) = -8\ V}$$

(b) Applying KCL at the $\boldsymbol{v_o}$ node,

$$i_o = \frac{v_o - v_-}{10} + \frac{v_o}{2} = (-0.8 - 4)\ \text{mA} = \boldsymbol{-4.8\ mA}$$

Example 05

Design an op-amp circuit with inputs $\boldsymbol{v_1}$, $\boldsymbol{v_2}$ and $\boldsymbol{v_3}$ such that, output voltage $\boldsymbol{v_o = -\frac{2}{5}v_1 - \frac{4}{5}v_2 - \frac{4}{3}v_3}$

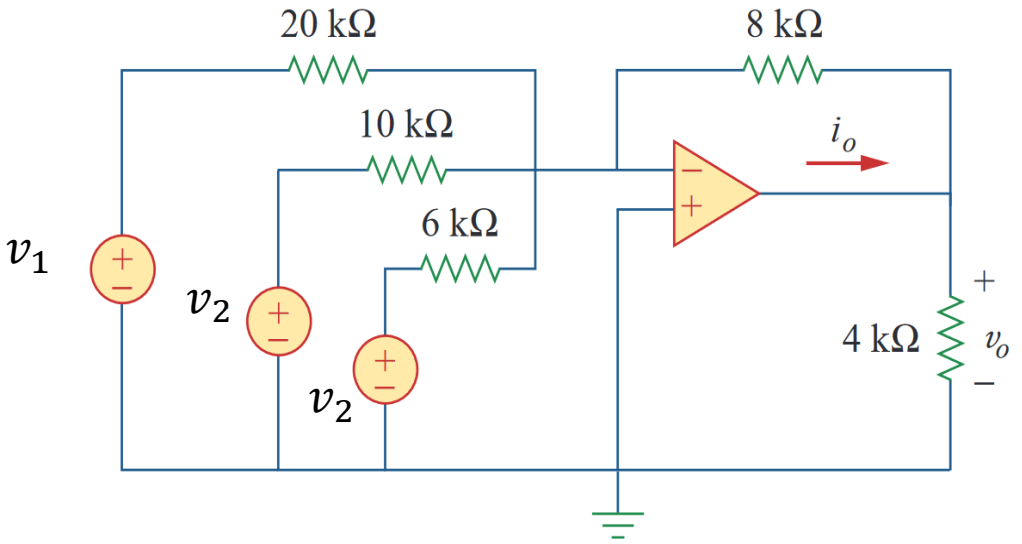
The equation for inverting summing amplifier is:

$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \cdots \frac{R_f}{R_n}v_n\right)$$

The numerators of all the coefficients of input voltages are same ($\boldsymbol{R_f}$). In the given problem, the numerator can be set to the LCM of 2 and 4, which is **8**.

$$\boldsymbol{v_o = -\left(\frac{2}{5}v_1 + \frac{4}{5}v_2 + \frac{4}{3}v_3\right)}$$

$$\boldsymbol{= -\left(\frac{8}{20}v_1 + \frac{8}{10}v_2 + \frac{8}{6}v_3\right)}$$



The Resulting Circuit

Example 05

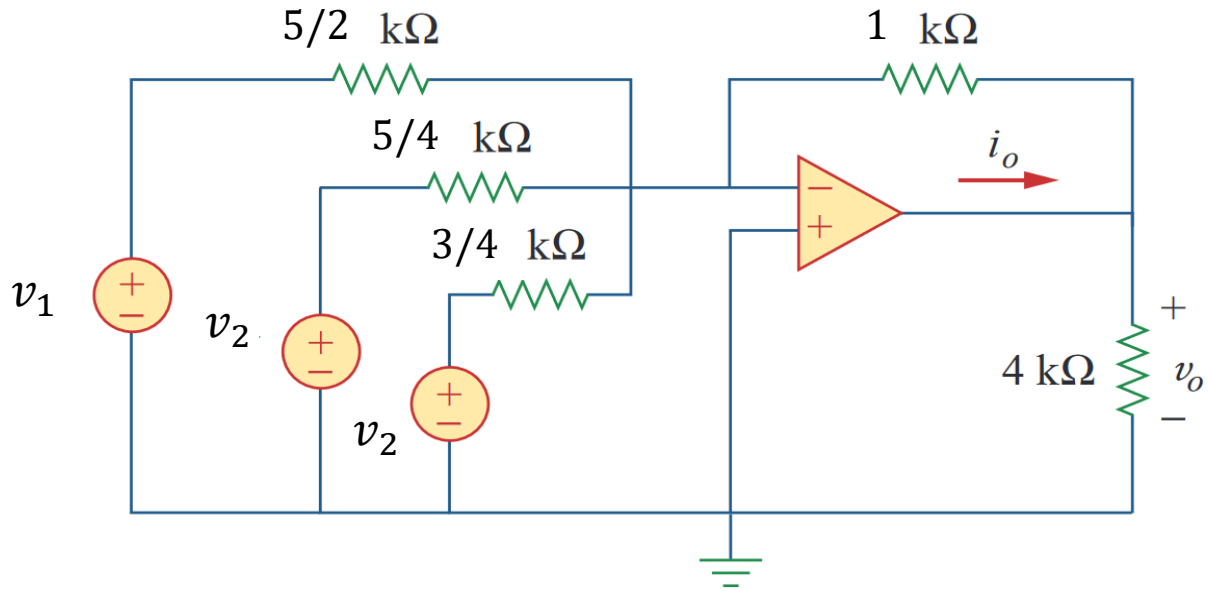
Design an op-amp circuit with inputs v_1, v_2 and v_3 such that, output voltage $v_o = -\frac{2}{5}v_1 - \frac{4}{5}v_2 - \frac{4}{3}v_3$

Easier Solution:

$$v_o = -\frac{2}{5}v_1 - \frac{4}{5}v_2 - \frac{4}{3}v_3$$

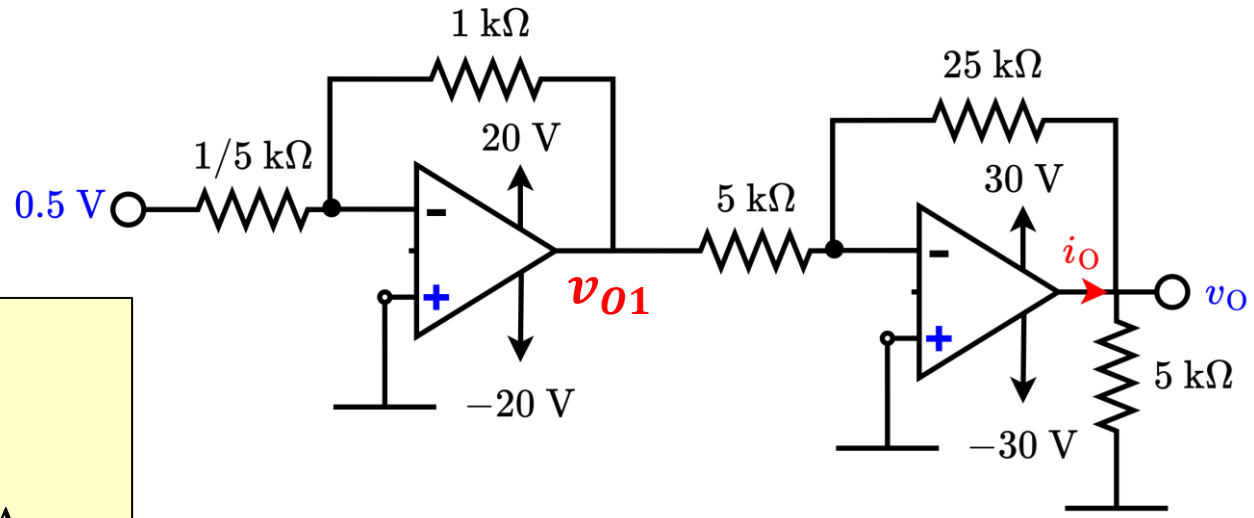
$$= -\frac{1}{5/2}v_1 - \frac{1}{5/4}v_2 - \frac{1}{3/4}v_3$$

$$= -\left(\frac{1}{5/2}v_1 + \frac{1}{5/4}v_2 + \frac{1}{3/4}v_3\right)$$



Example 06

$v_I = 0.5$ V. Calculate the Output voltage $\boldsymbol{v_o}$ and Output current $\boldsymbol{i_o}$.



The first Op-Amp is an Inverting Amplifier,

$$\boldsymbol{v_{O1} = -\frac{R_f}{R_i} \times v_{in} = -\frac{1}{1/5} \times 0.5\ \text{V} = -2.5\ \text{V}}$$

Which is within the saturation limit of the 1st Op-Amp.

The output from the first Op-Amp = The input for the second Op-Amp. The 2nd Op-Amp is also an Inverting Amplifier.

$$\boldsymbol{v_o = -\frac{R_f}{R_i} \times v_{o1} = -\frac{25}{5} \times -2.5 = 12.5\ \text{V}}$$

Which is within the saturation limit of the 2nd Op-Amp.

Applying KCL at the $\boldsymbol{v_o}$ node,

$$\boldsymbol{i_o} = \frac{v_o}{5} + \frac{v_o}{25} = (2.5 + 0.5)\ \text{mA}$$

$$= \boldsymbol{3\ \text{mA}}$$

Non-Inverting Amplifier

A circuit that amplifies the Input V_{in} to the Output V_o and preserves its polarity (doesn't invert).

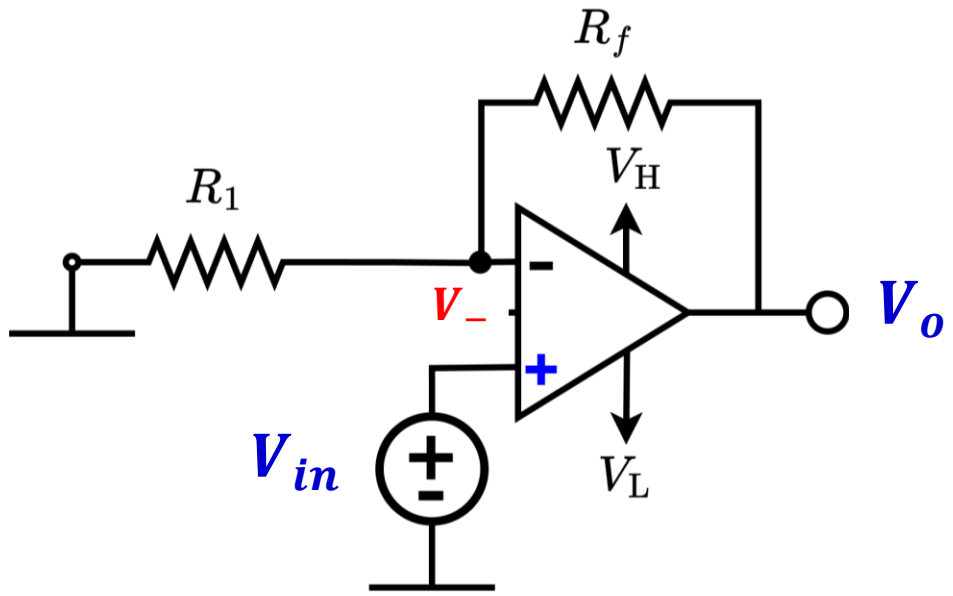
Input is given at the V_+ node, $V_+ = V_{in}$
Using Virtual Short, $V_- = V_+ = V_{in}$

Applying KCL at V_- node,

$$\frac{0 - V_-}{R_1} = I_- + \frac{V_- - V_o}{R_f}$$

$$\Rightarrow \frac{V_o}{R_f} = \left(\frac{V_{in}}{R_1} + \frac{V_{in}}{R_f}\right) \quad [I_- = 0]$$

$$\Rightarrow \boldsymbol{V_o = \left(1 + \frac{R_f}{R_1}\right) \times V_{in}}$$



Comparing the final equation with,
$V_o = Gain \times V_{in}$

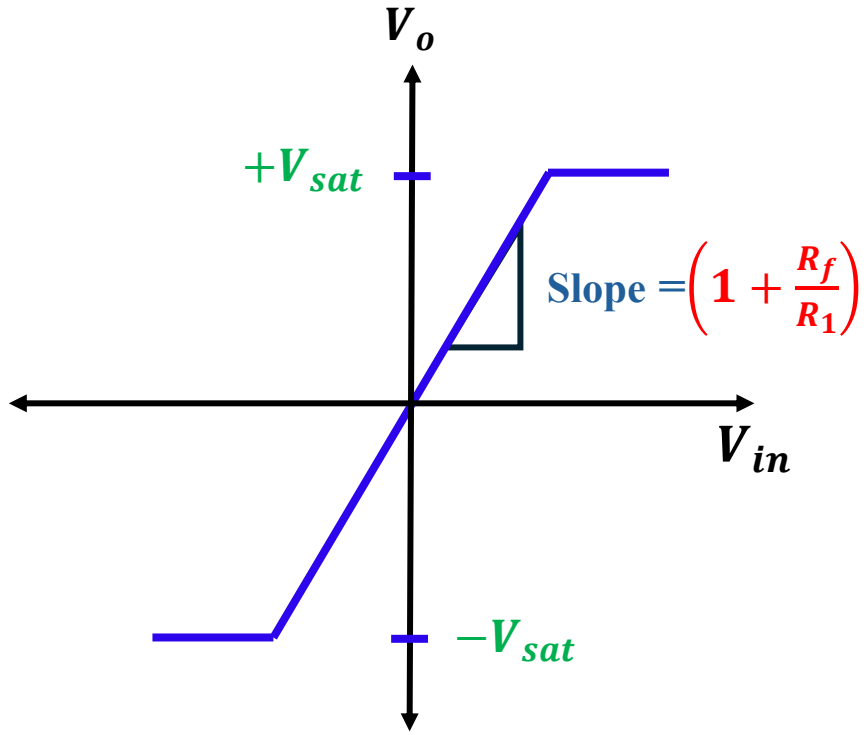
→ $Gain = \left(1 + \frac{R_f}{R_1}\right)$ →

Gain is **Positive** and always **>1,** depending on R_f and R_1

Non-Inverting Amplifier

- Gain for Non-Inverting Amplifier $= \left(1 + \frac{R_f}{R_1}\right)$
- Hence, the **VTC** plot is a straight line with a **Positive slope.**
- The polarity of Input and Output doesn't change.
- We can adjust the **Gain** by changing R_f and R_I, but **cannot achieve a Positive Gain < 1**, such as a gain of 0.5 cannot be achieved by this amplifier.
- However, V_o can never cross the saturation limit.

$$-V_{sat} \leq V_o \leq +V_{sat}$$



VTC of a Non-Inverting Amplifier

Example 7

Calculate the Output voltage V_o and Output current I_o.

Apply KCL at V_+ node,

$$\frac{3 - V_+}{2} = 0 + \frac{V_+}{4}$$

$$\Rightarrow V_+ = 2V$$

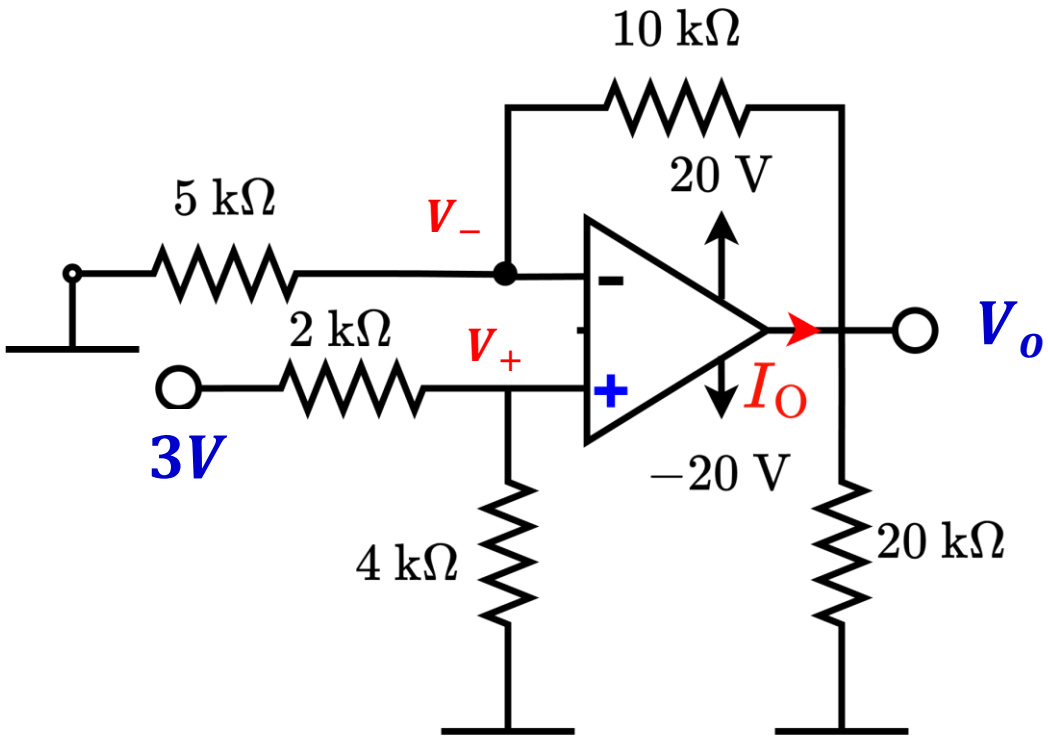
The remaining circuit is a Non-Inverting Amplifier,

$$V_o = \left(1 + \frac{10}{5}\right) \times V_+ = 6\ \text{V}$$

Alternatively, we can also apply Virtual Short and KCL at V_- node to find V_o as well.

Now, applying KCL at V_o node,

$$I_o = \frac{V_o}{20} + \frac{V_o - V_-}{10} = \left(\frac{6}{20} + \frac{6-2}{10}\right) = 0.7\ \text{mA}$$



Weighted Subtractor

A circuit that subtracts two weighted input signals (V_1 from V_2) in the Output V_o.

Apply KCL at $V_{(+)}$ node,

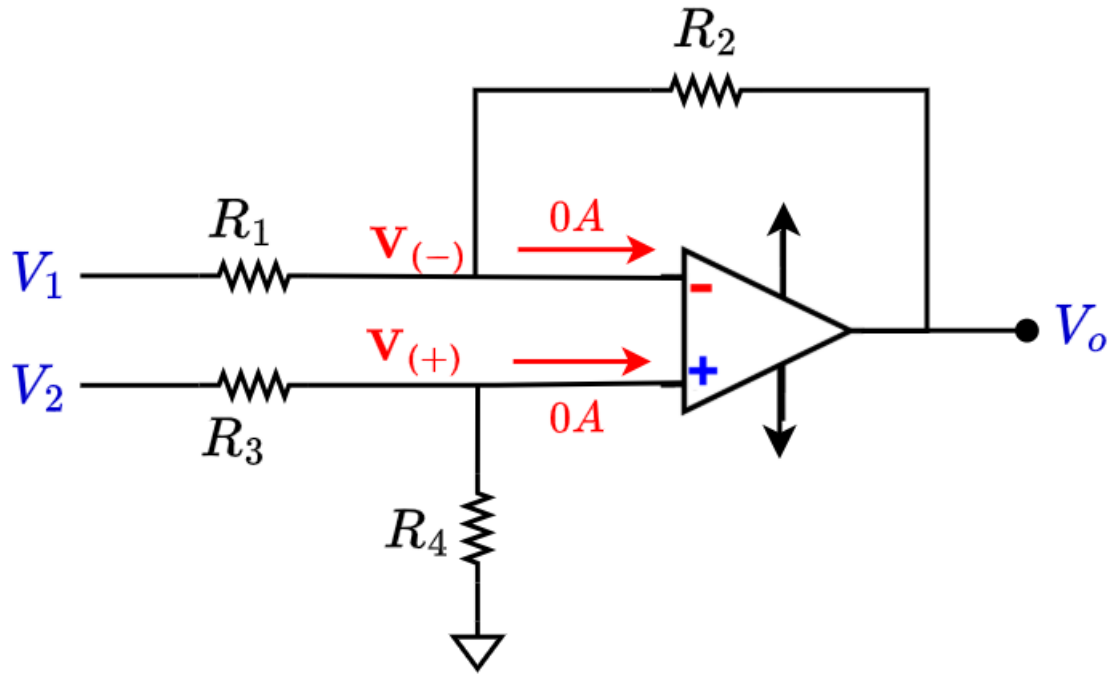
$$\frac{V_2 - V_{(+)}}{R_3} = 0 + \frac{V_{(+)}}{R_4}$$

$$\Rightarrow V_{(+)} = \frac{R_4}{R_3 + R_4} \times V_2 \; --(1)$$

Apply KCL at $V_{(-)}$ node,

$$\frac{V_1 - V_{(-)}}{R_1} = 0 + \frac{V_{(-)} - V_o}{R_2}$$

$$\Rightarrow V_o = V_{(-)}\left(1 + \frac{R_2}{R_1}\right) - \frac{R_2}{R_1} V_1 \; --(2)$$



Weighted Subtractor

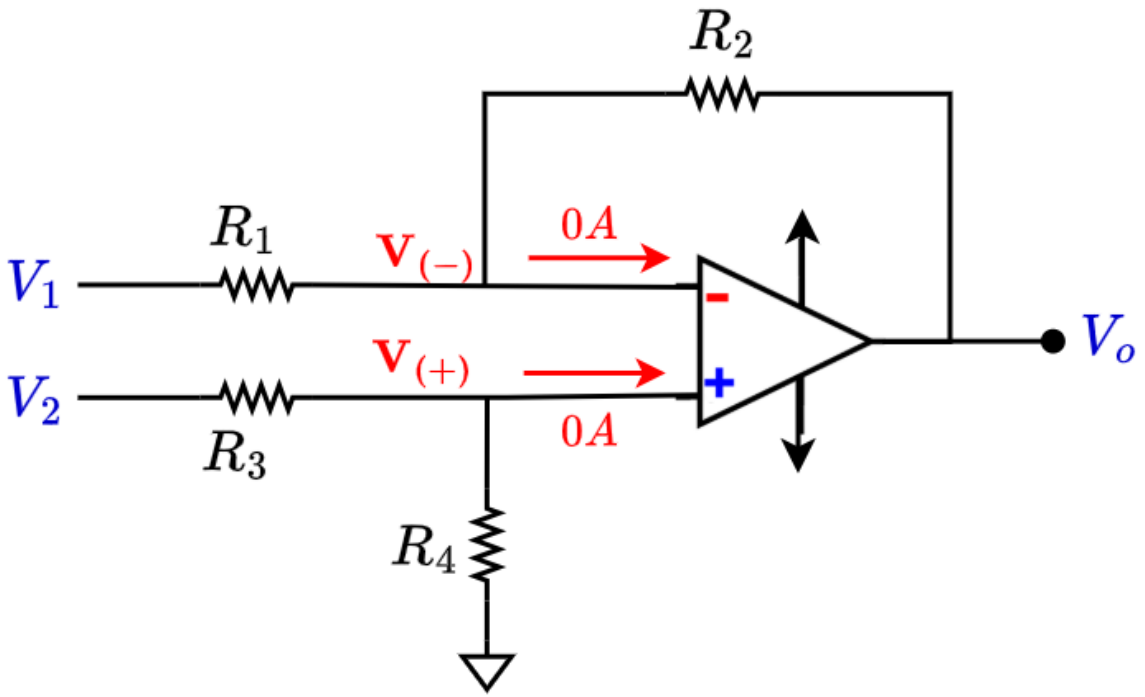
From Virtual Short, $V_{(+)} = V_{(-)}$

Hence, substituting $V_{(+)}$ from Eq. (1) into Eq. (2):

$$V_o = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right)V_2 - \frac{R_2}{R_1}V_1$$

$$= \left(1 + \frac{R_2}{R_1}\right)\left(\frac{1}{1 + \frac{R_3}{R_4}}\right)V_2 - \frac{R_2}{R_1}V_1$$

$$V_o = \frac{1 + (R_2/R_1)}{1 + (R_3/R_4)}V_2 - \frac{R_2}{R_1}V_1$$



Each input is multiplied by a **factor** and the weighted products are **subtracted**. The input connected to the Inverting terminal (V_1) is subtracted from the input connected to the Non-inverting terminal (V_2).

Simple Subtractor

A circuit that simply subtracts two input signals (V_1 from V_2) in the Output V_o without weighting.

From the weighted subtractor circuit,

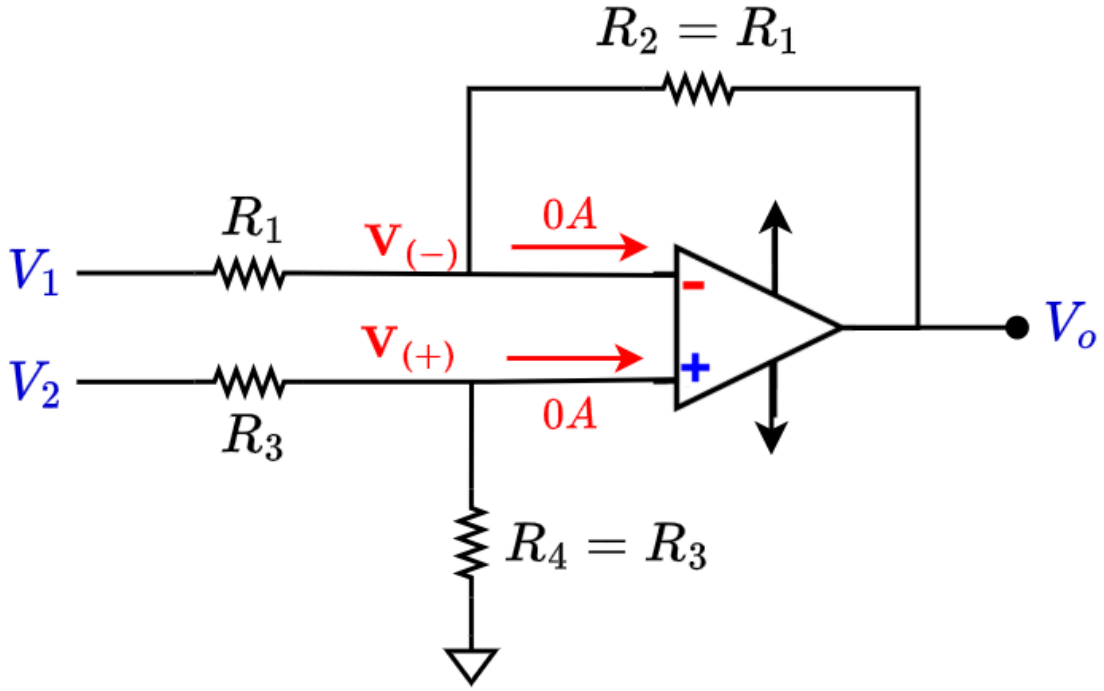
$$V_o = \frac{1 + (R_2/R_1)}{1 + (R_3/R_4)} V_2 - \frac{R_2}{R_1} V_1$$

If we set $R_2 = R_1$ and $R_4 = R_3$;

$$V_o = \frac{1 + (R_1/R_1)}{1 + (R_3/R_3)} V_2 - \frac{R_1}{R_1} V_1$$

$$or, V_o = \frac{1+1}{1+1} V_2 - V_1$$

$$or, V_o = V_2 - V_1$$



The input connected to the Inverting terminal (V_1) is simply subtracted from the input connected to the Non-inverting terminal (V_2), without any weighting factor.

Example 8

Solve the following circuit to find Output V_o.

From the weighted subtractor circuit, we find:

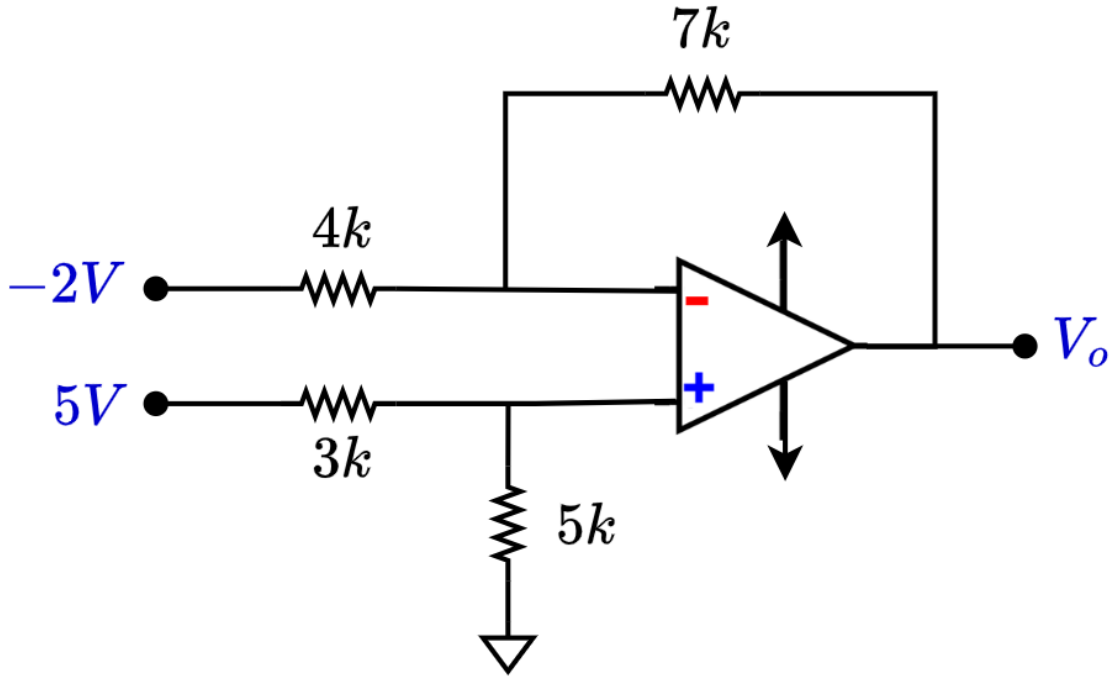
$$R_1 = 4k; R_2 = 7k; R_3 = 3k; R_4 = 5k$$

Hence, using formula:

$$V_o = \frac{1 + (R_2/R_1)}{1 + (R_3/R_4)} V_2 - \frac{R_2}{R_1} V_1$$

$$= \frac{1 + (7/4)}{1 + (3/5)} 5 - \frac{7}{4}(-2)$$

$$= 12.093\ V$$



You can also solve this problem using the general steps to solve any Op-Amp Closed Loop circuits, as shown in Example 01.

Inverting Differentiator

A circuit that performs Differentiation on the Input V_{in} to the Output V_o.

The V_+ node is grounded, $V_+ = 0V$
Using Virtual Short, $V_- = V_+ = 0V$

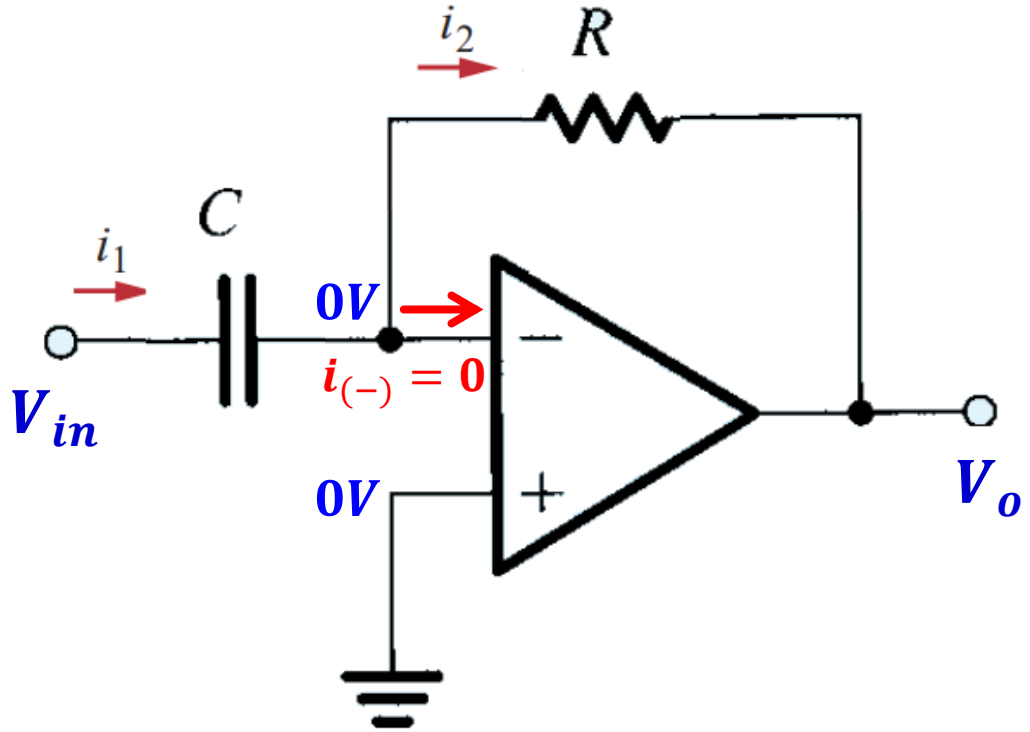
Applying KCL at V_- node,

$$i_1 = i_{(-)} + i_2$$

$$C\frac{d(V_{in}-0)}{dt} = \frac{0 - V_O}{R}$$

$$\Rightarrow -\frac{V_o}{R} = C\frac{dV_{in}}{dt}$$

$$\Rightarrow V_O = -RC\frac{dV_{in}}{dt}$$



The input V_{in} gets **Differentiated** and multiplied by a **factor $(-RC)$** in the Output V_o.

Inverting Integrator

A circuit that performs Integration on the Input V_{in} to the Output V_o.

The V_+ node is grounded, $V_+ = 0V$
Using Virtual Short, $V_- = V_+ = 0V$

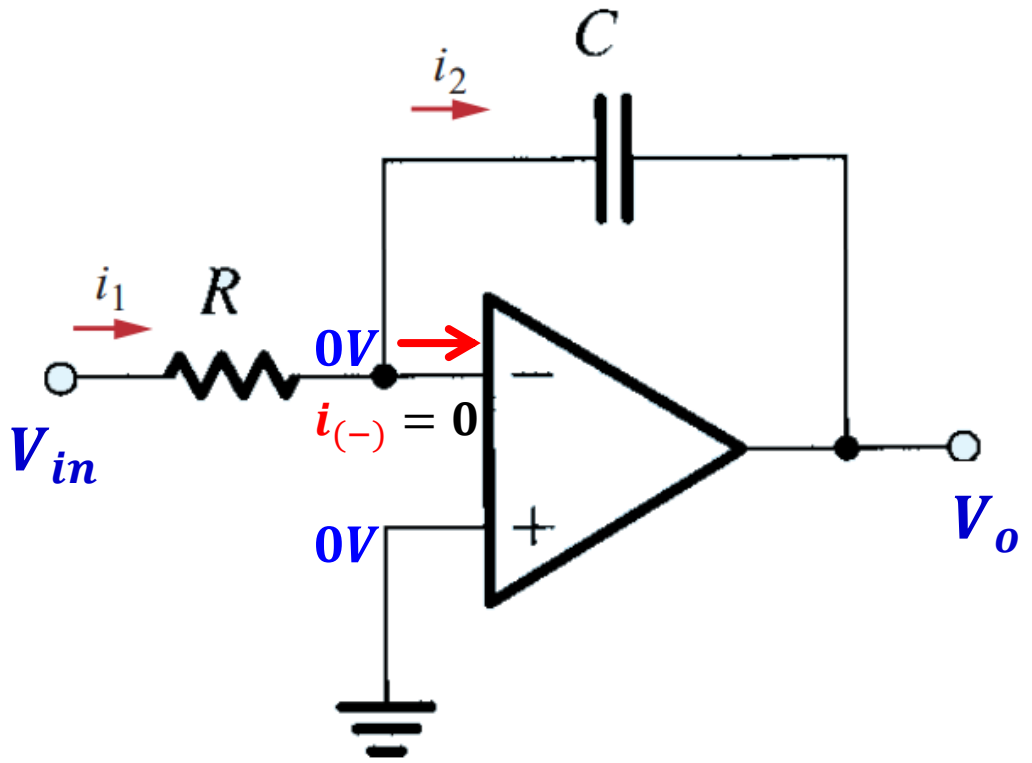
Applying KCL at V_- node,

$$i_1 = i_{(-)} + i_2$$

$$\frac{V_{in} - 0}{R} = C\frac{d(0 - V_o)}{dt}$$

$$\Rightarrow \frac{V_{in}}{R} = -C\frac{dV_o}{dt}$$

$$\Rightarrow V_O = -\frac{1}{RC}\int V_{in}(t)dt$$



The input V_{in} gets **Integrated** and multiplied by a **factor** $(-\frac{1}{RC})$ in the Output V_o.

Example 9

Observe the following Figure. If $V_{in} = 5\sin(6t)$, Find the value of V_0

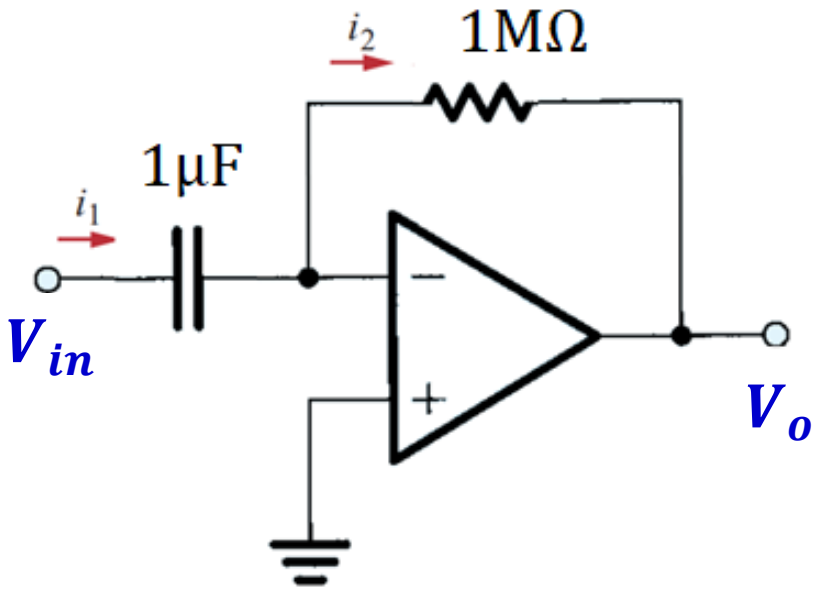
This is a **Differentiator**.

$$v_o = -RC\frac{\mathbf{d}V_{in}}{\mathbf{d}t}$$

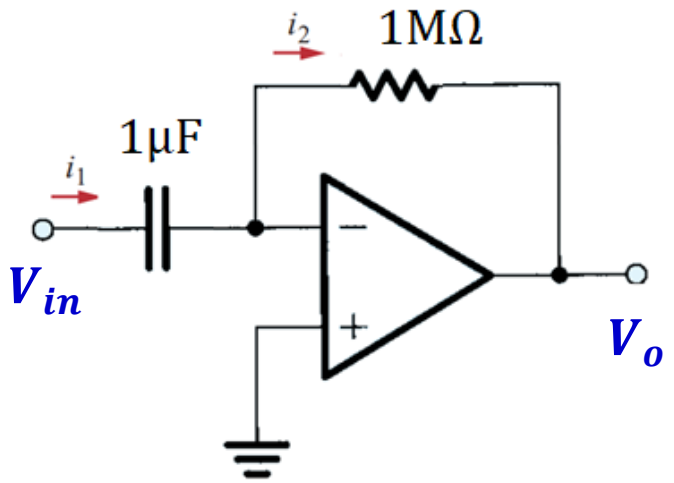
$$= -(1 \times 10^6) \cdot (1 \times 10^{-6}) \times \frac{\mathbf{d}(5 \cdot \sin(6t))}{\mathbf{d}t}$$

$$= -1 \times (5 \times 6\cos(6t))$$

$$= \mathbf{-30\cos(6t)}$$



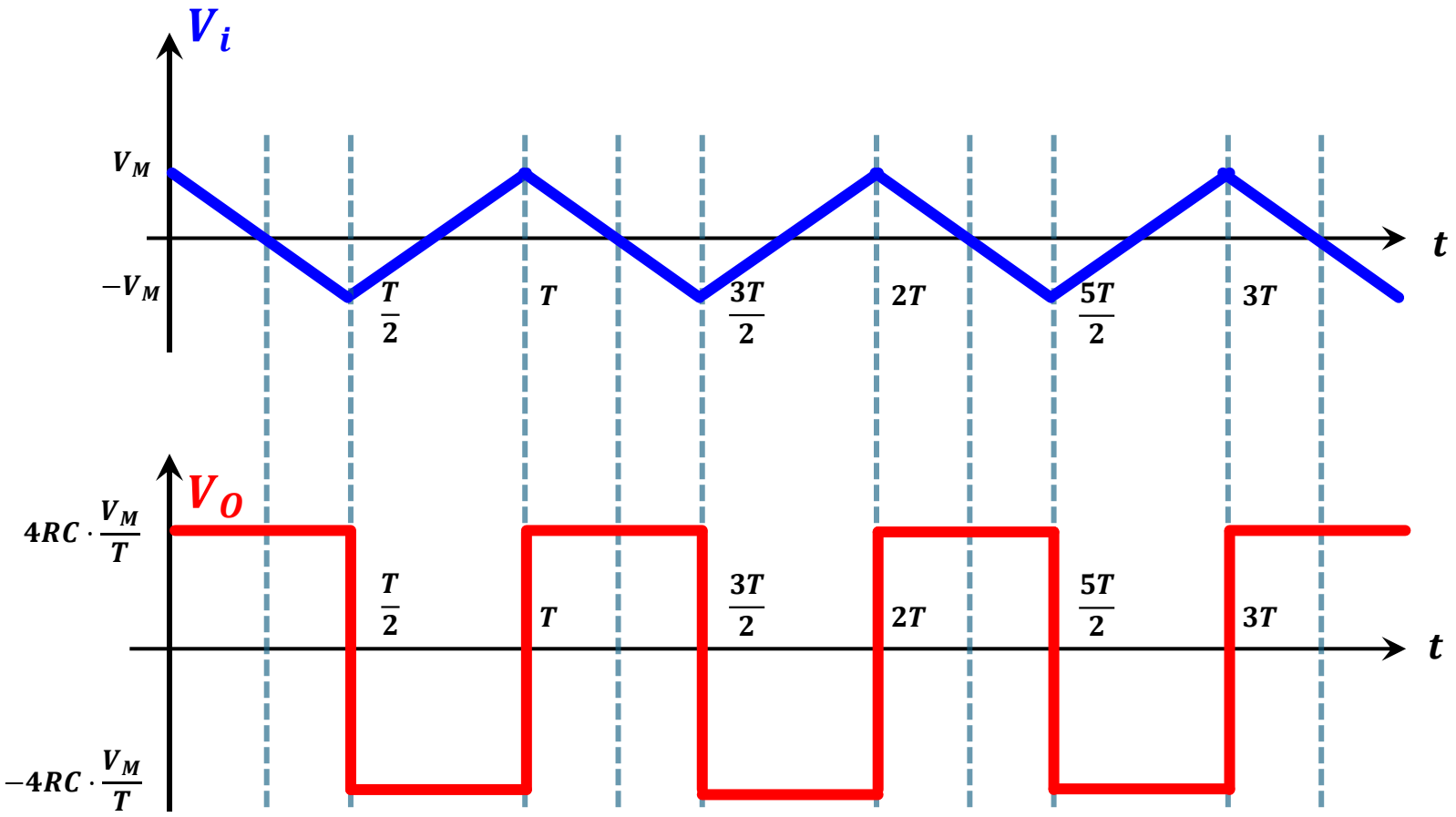
Example 10



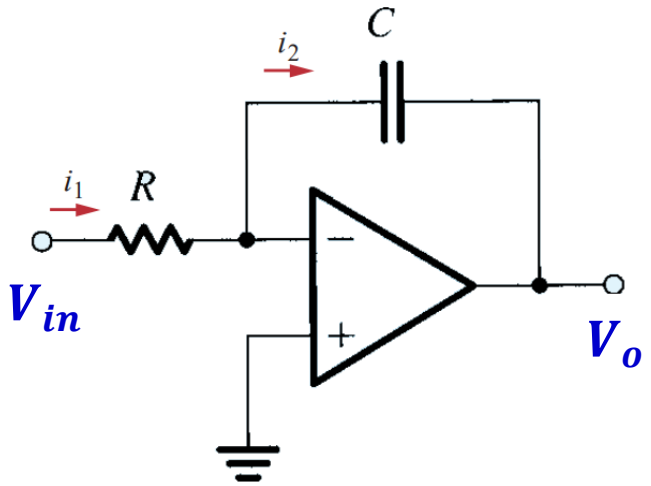
Differentiation of V_{in} is basically slope of V_{in}.

$$\left|\frac{dV_{in}}{dt}\right| = \frac{V_M - (-V_M)}{T/2} = \frac{4V_M}{T}$$

$$V_O = -RC\frac{dV_{in}}{dt} = -RC\frac{4V_M}{T}$$



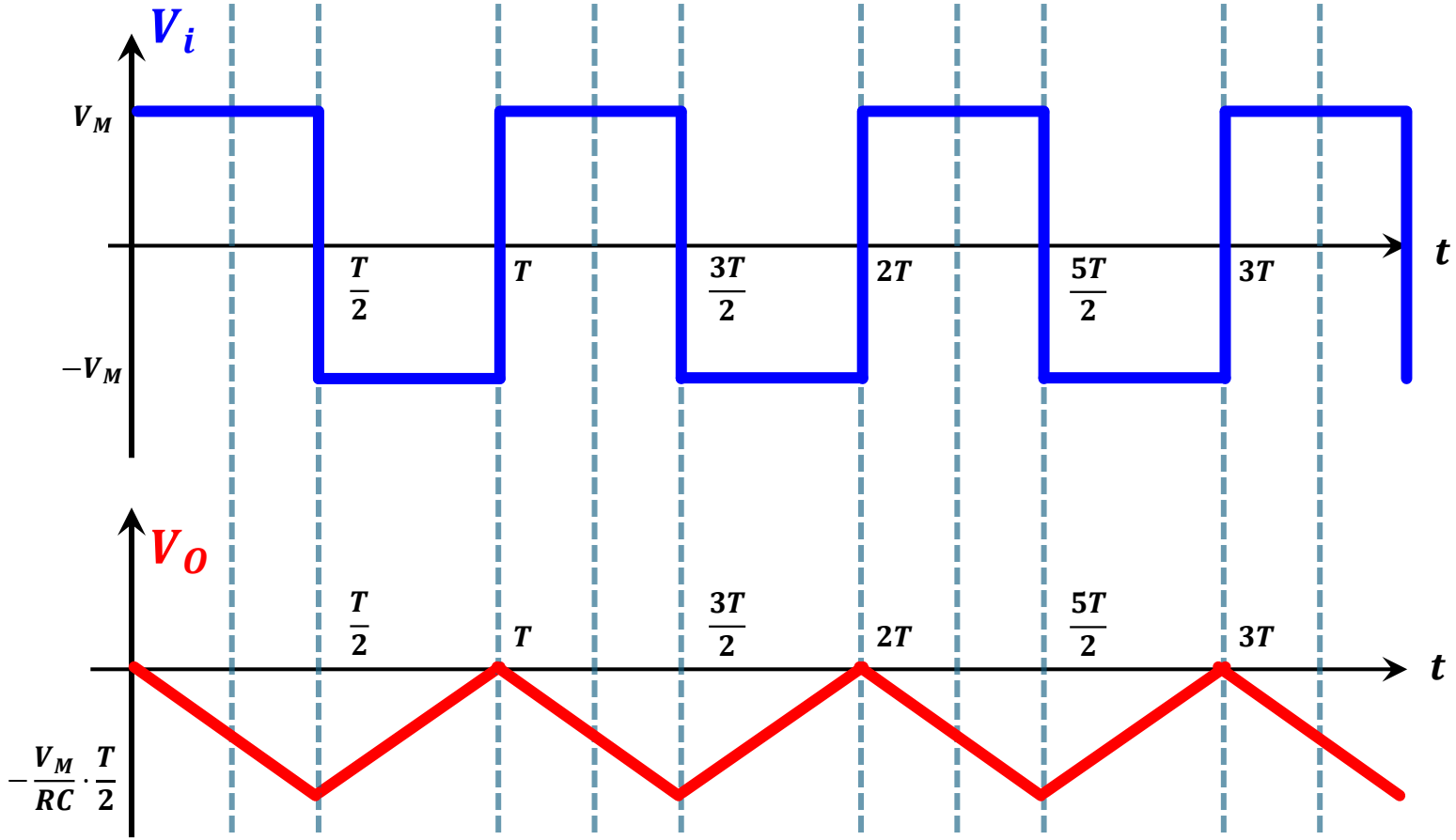
Example 11



Integration of V_{in} is basically the Area of V_{in} with time.

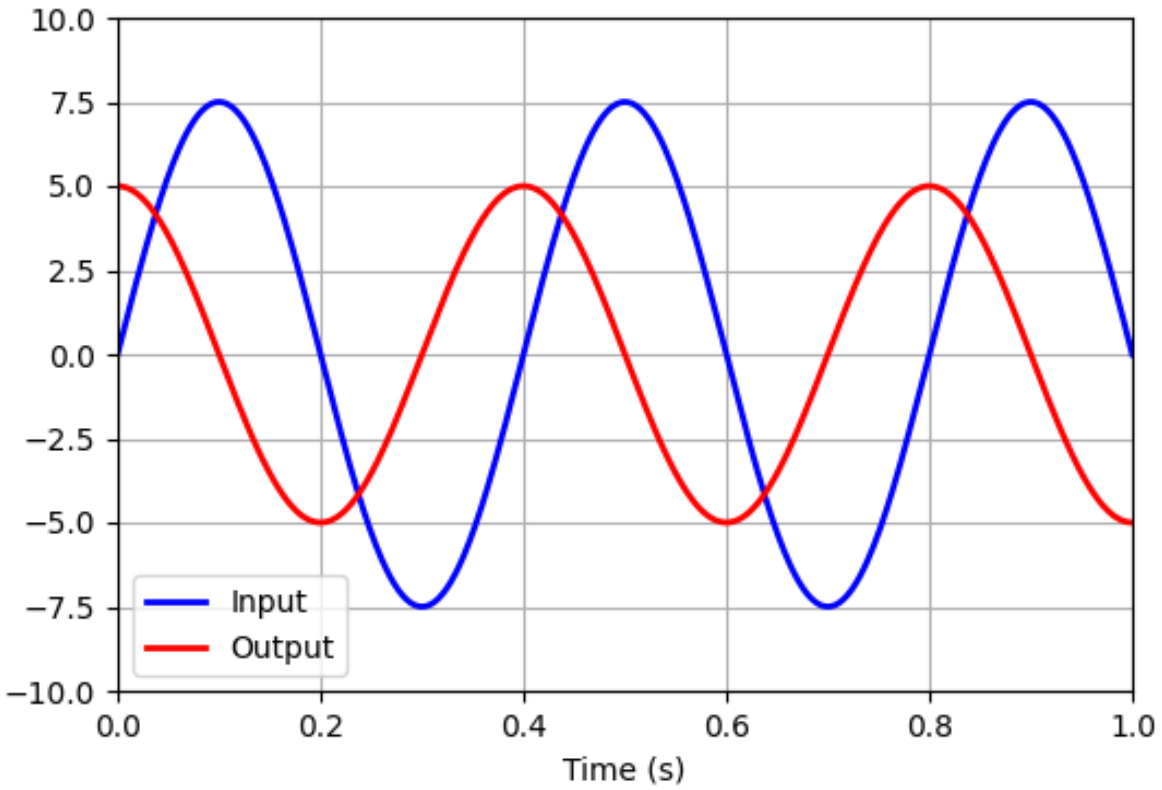
$$\int_0^t V_{in}\,dt = V_M \cdot \frac{T}{2}$$

$$V_O = -\frac{1}{RC}\int V_{in}(t)dt = -\frac{V_M}{RC}\cdot\frac{T}{2}$$



Example 12

Design an Op-Amp circuit that can transform input signal to output signal as shown in the figure.



Observe that,

$$V_{in} = 7.5\sin(\omega t)$$
$$V_{out} = 5\cos(\omega t)$$

Since, an input sine wave is transformed into an output cosine wave, you have to try Differentiator or Integrator circuits.

Hint: Carefully think which one (Differentiator or Integrator circuit) takes a positive sine wave and returns a positive cosine wave!!

Implementing Operational Functions

- $\boldsymbol{f = -4x + 5y}$
- $\boldsymbol{f = -7x + \frac{dy}{dt}}$
- $\boldsymbol{f = -5\frac{dy}{dt} + 2\int x\,dt - 4z}$
- $\boldsymbol{f = \frac{d^2x}{dt^2} + \frac{5}{3}\int\int y\;dtdt}$

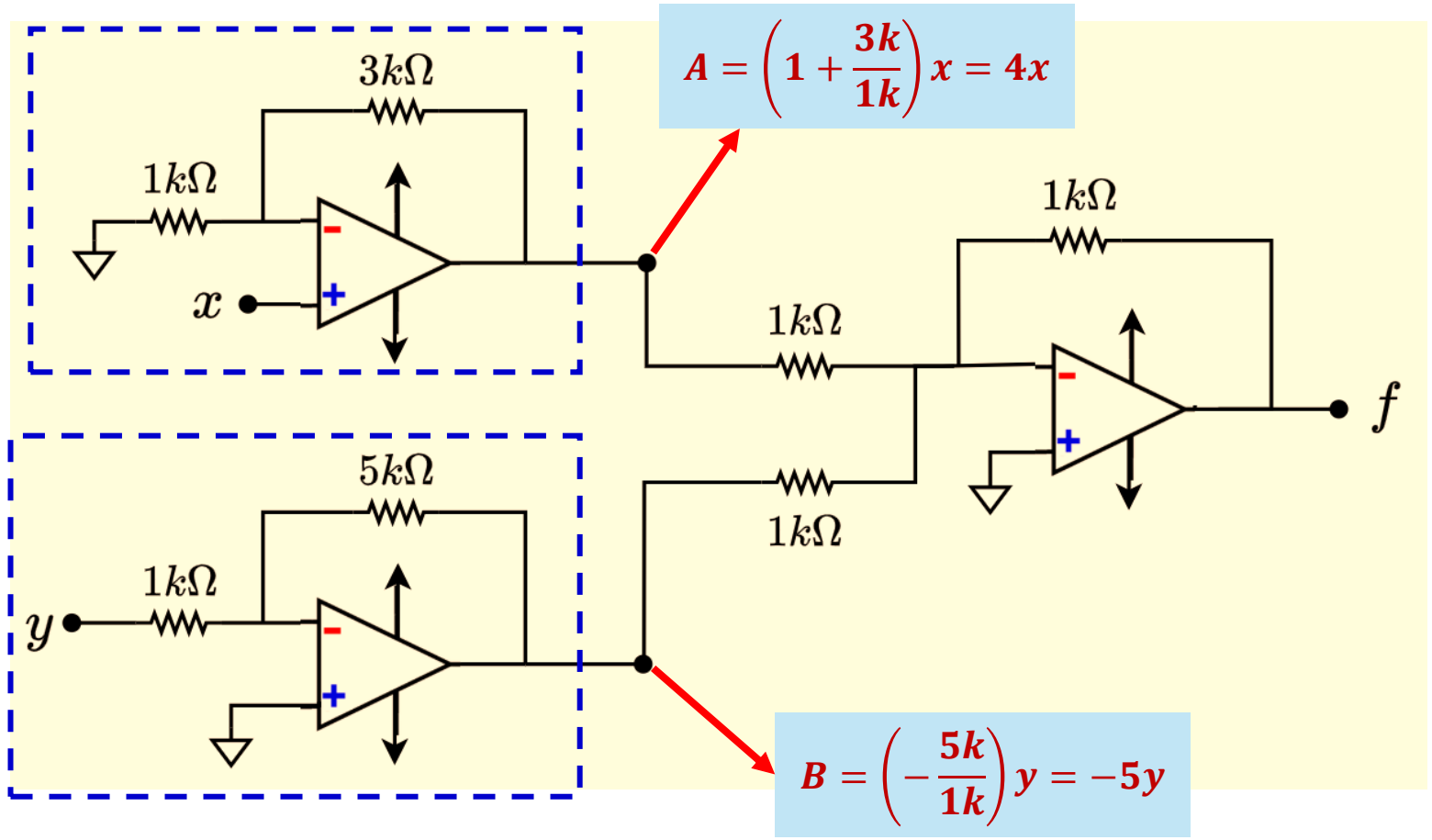
We can utilize the different Amplifiers discussed in the previous sections to implement these mathematical operations.

- $f = -\frac{1}{3}\int x \cdot dt + 2\ln y + 4z$
- $f = -3\frac{dx}{dt} + 2\exp(y) + 4z$
- $f = xy/z$

We need "Diode" to implement Logarithmic, Exponential, Multiplication, and Division operations, which will be covered after Diode is analyzed in the next chapter.

Operational Functions: Example 01

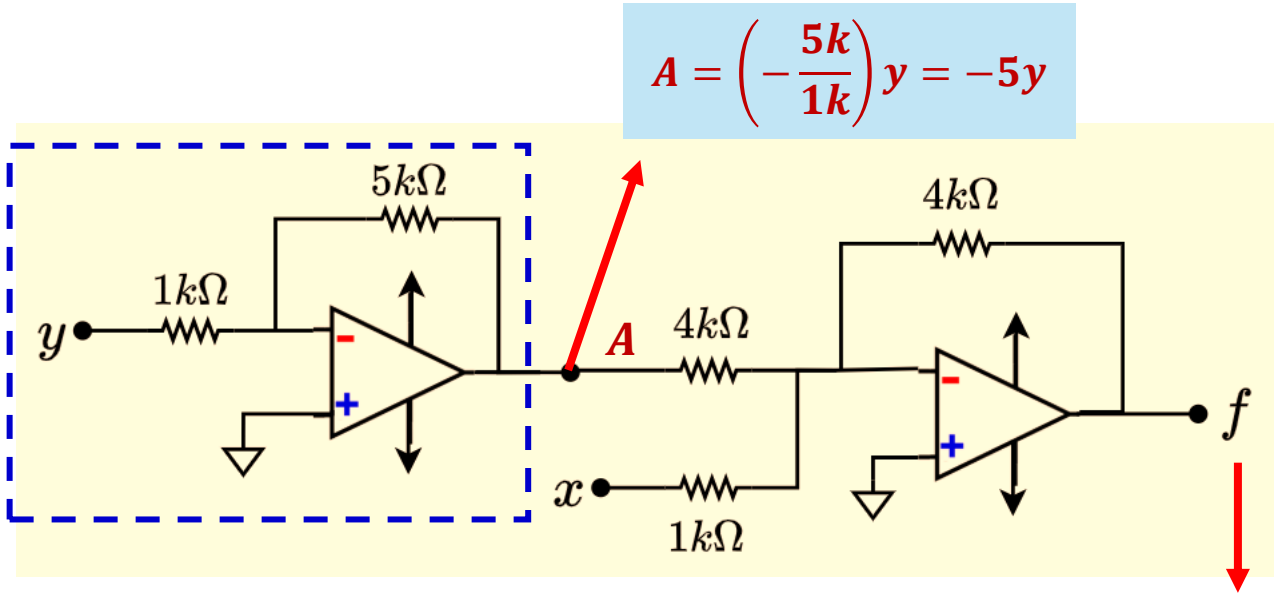
$$f = -4x + 5y = -(4x - 5y) = -(A + B) \qquad \text{Where, } A = 4x \text{ and } B = -5y$$

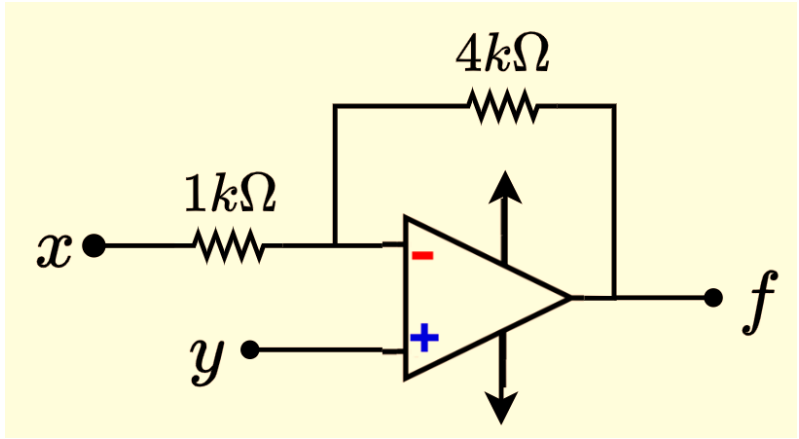


Operational Functions: Example 13

Alternative ways to implement the same function,

$f = -4x + 5y = -(4x + (-5y)) = -\left(\frac{4}{1}x + \frac{4}{4}A\right)$; where $A = -5y$





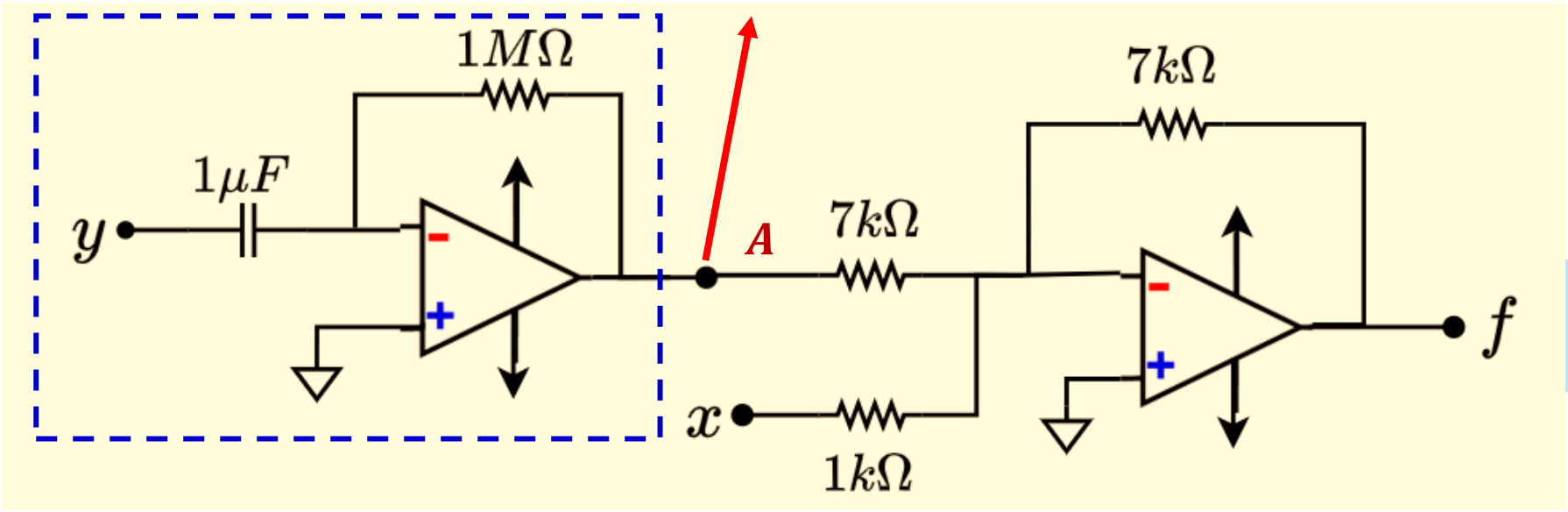
$f = f_x + f_y = -\frac{4k}{1k}x + \left(1 + \frac{4k}{1k}\right)y$

$f = -4x + 5y$

Operational Functions: Example 14

$$f = -7x + \frac{dy}{dt} = -\left(7x - \frac{dy}{dt}\right) = -\left(\frac{7}{1}x + \frac{7}{7}A\right); \textit{where } A = -\frac{dy}{dt}$$

$$A = -1\mu F \times 1M\Omega \times \frac{dy}{dt}$$



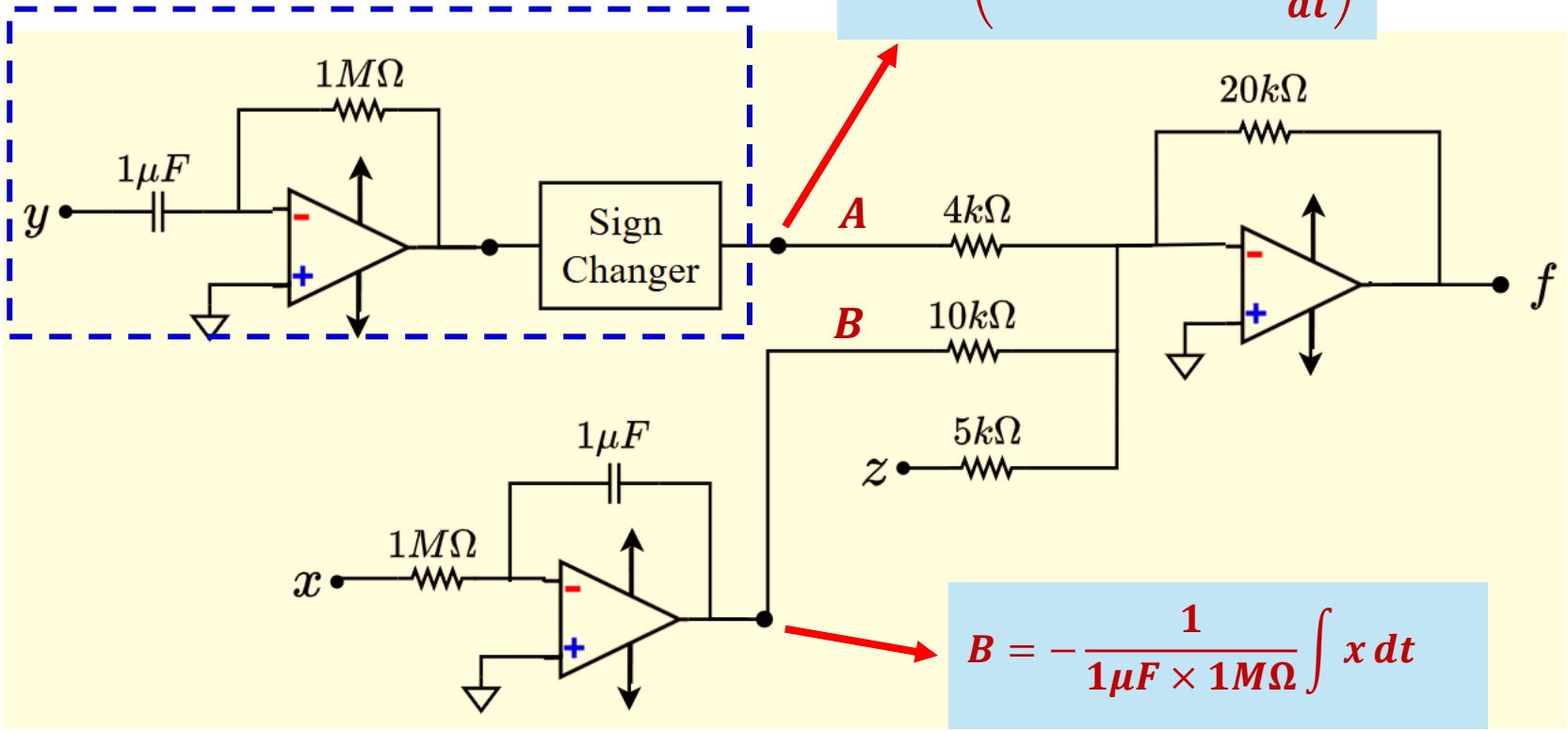
$$= -\left(\frac{7k}{1k}x + \frac{7k}{7k}A\right)$$

Operational Functions: Example 15

$$f = -5\frac{dy}{dt} + 2\int x\,dt - 4z = -\left(5\frac{dy}{dt} - 2\int x\,dt + 4z\right) = -\left(\frac{20}{4}A + \frac{20}{10}B + \frac{20}{5}z\right)$$

Where, $A = \frac{dy}{dt}$ and $B = -\int x\,dt$

$$A = -\left(-1\mu F \times 1M\Omega \times \frac{dy}{dt}\right)$$



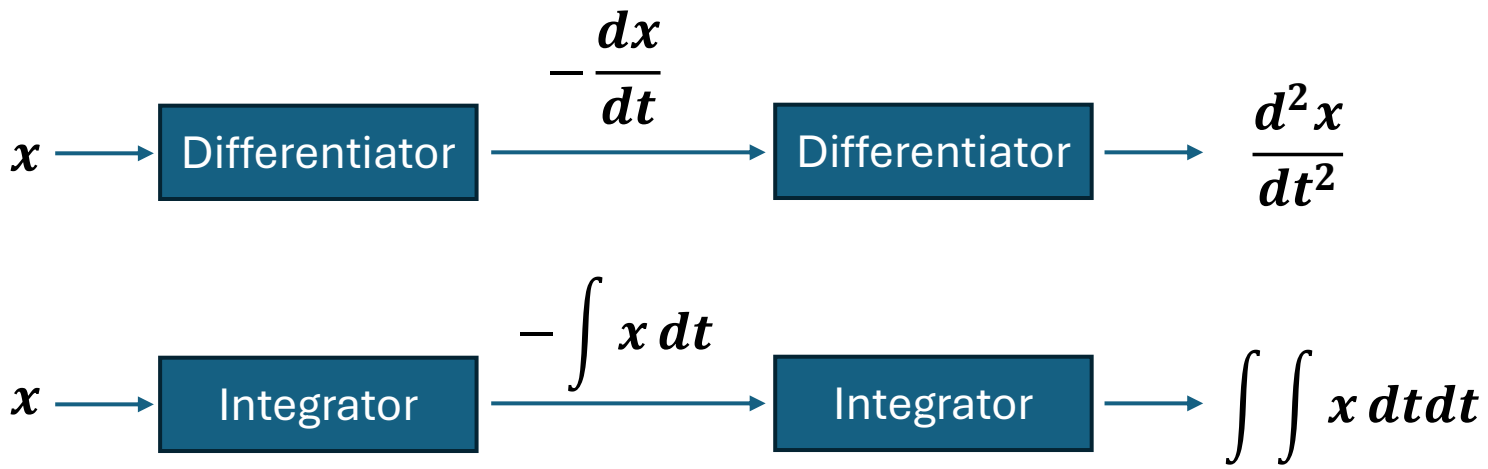
$$B = -\frac{1}{1\mu F \times 1M\Omega}\int x\,dt$$

$$= -\left(\frac{20}{4}A + \frac{20}{10}B + \frac{20}{5}z\right)$$

Operational Functions: Example 16

$$f = \frac{d^2x}{dt^2} + \frac{5}{3}\int\int y\,dtdt$$

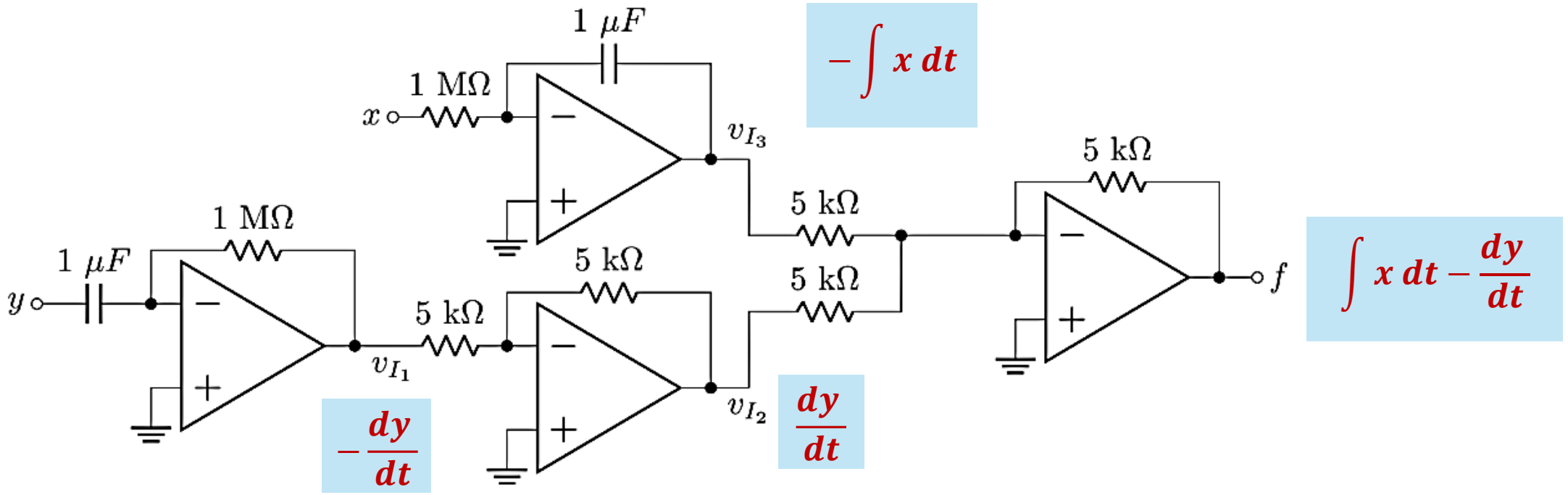
Hints: In case of Double Derivative or Double Integration, you need to cascade subsequent differentiator or integrator circuits. Similar approach can be followed for ‘n’-th Derivative or ‘n’-th Integral.



Solve the rest by yourself, following the previous examples.

Example 17: Expression Calculation

Analyze the circuit below to **find** an expression of f in terms of inputs x and y.



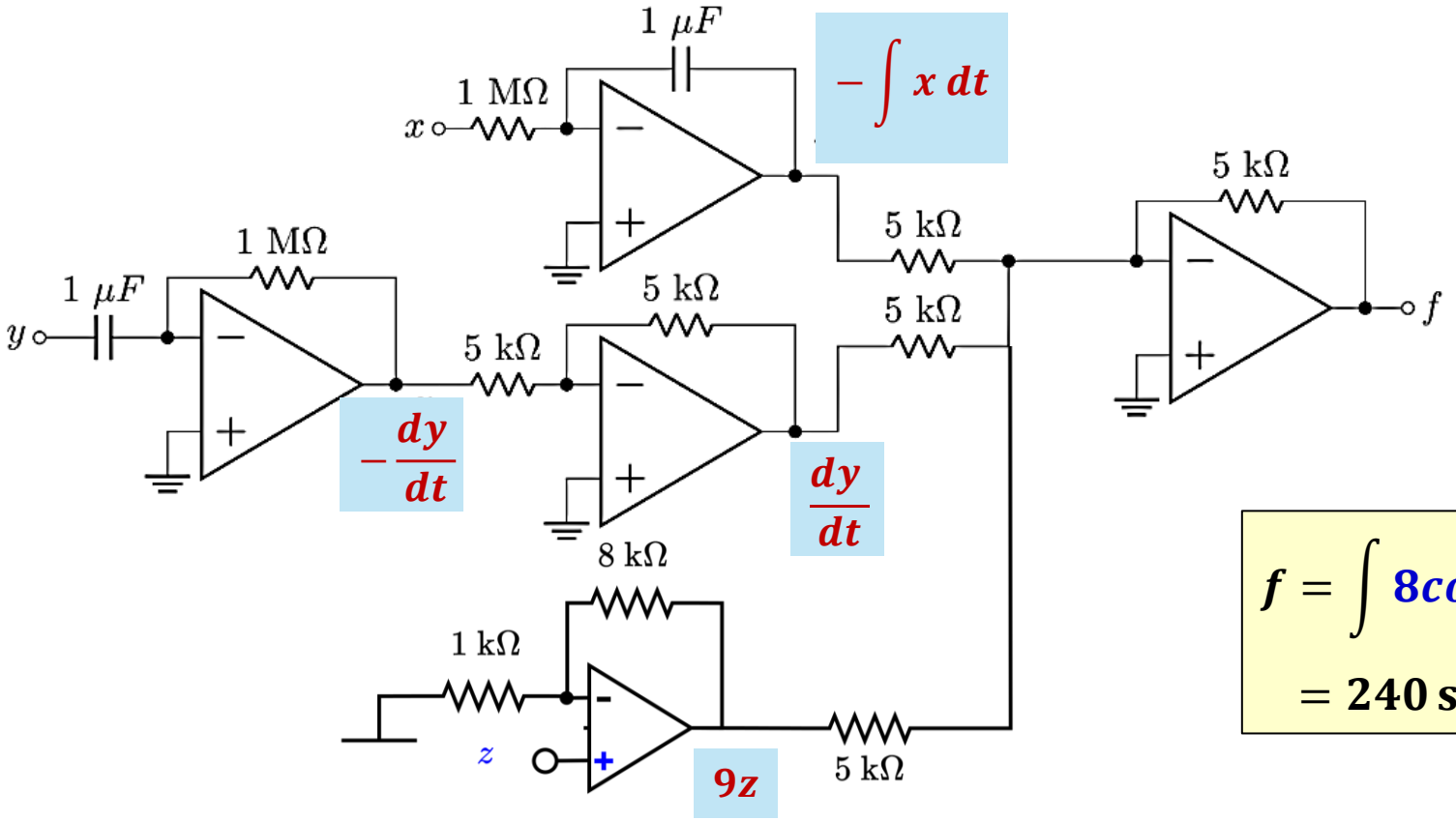
Solution:

$$v_{I_1} = -RC\frac{dy}{dt}; \quad v_{I_2} = -v_{I_1} = \frac{dy}{dt}; \quad v_{I_3} = -\frac{1}{RC}\int x\,dt; \quad f = -\left(\frac{5}{5}v_{I_2} + \frac{5}{5}v_{I_3}\right) = -(v_{I_2} + v_{I_3})$$

Example 18: Expression Calculation

Analyze the circuit below to **find** an expression of f in terms of inputs x, y and z.

If $x = 8\cos(30t)$ V and $y = 3\sin(30t)$ V and $z = 1$ V



$$= \int x\,dt - \frac{dy}{dt} - 9z$$

$$f = \int 8\cos(30t)dt - \frac{d}{dt}(3\sin(30t)) - 9 \times 1$$

$$= 240\sin(30t) - 90\cos(30t) - 9 \text{ V}$$